

**COUNTING THE UNCOUNTED: EXPLORING MATHEMATICAL LITERACY
LEVELS AMONG KINDERGARTENERS IN NEW CABALAN ELEMENTARY
SCHOOL**

A Research
Presented to the Faculty of the
College of Education, Arts and Sciences
Gordon College
Olongapo City

In Partial Fulfillment of the
Requirements for the Degree
Early Childhood Education

Garganta, Guienn
Librero, Alyssa Rica
Mesina, Donna May
Villafania, Georgia Victoria

May 14, 2025



APPROVAL SHEET

This Research entitled **"COUNTING THE UNCOUNTED: EXPLORING MATHEMATICAL LITERACY LEVELS AMONG KINDERGARTENERS IN NEW CABALAN ELEMENTARY SCHOOL"** prepared and submitted by **Guienn Garganta, Alyssa Rica Librero, Donna May Mesina and Georgia Victoria Villafania** in partial fulfillment of the requirements for the degree of **BACHELOR OF EARLY CHILDHOOD EDUCATION** has been examined and is recommended for acceptance for approval for ORAL EXAMINATION.

Rachelle Ann D. Ignacio, LPT
Adviser

PANEL OF EXAMINERS

Approved by the Committee on Oral Examination with a grade of ____.

Ms. Anna Theresa Amarille, LPT, MASE
Assistant Dean, College of Education, Arts and Sciences
Chairman

Ms. Keana Denisse F. David, LPT
Member

Accepted and approved in partial fulfillment of the requirements for the degree BACHELOR OF EARLY CHILDHOOD EDUCATION.

May 14, 2026
Date of Oral Examination

Eder Magsayo, LPT, PhD
Dean, College of Education, Arts and Sciences



ABSTRACT

This study started with a simple question: how are our kindergarteners doing in early math, and how much do they grow through regular classroom instruction. The researchers focused on 42 pupils at New Cabalan Elementary School in Olongapo City, looking at their skills in color and shape recognition, number sense, and readiness for basic operations.

Using a descriptive-comparative design based on Clements and Sarama's Learning Trajectories Approach, the researchers assessed the children twice, by using pretest and posttest. No special intervention was introduced but relied in the teacher's daily classroom intervention setting. The goal was simply to see what they already knew and how they progressed through everyday teaching.

The findings were reassuring. Most children already had solid foundational skills from the start. During the post test, their scores improved significantly, especially among those who began with the biggest challenges.

When the researchers looked at age and sex, no major differences appeared. Boys and girls, five and six-year-olds, all grew similarly.

In short, this study offers a hopeful look at early math development in a real classroom.

Keywords: Mathematical literacy, Kindergarten, Early numeracy, Number recognition, Descriptive-comparative design



ACKNOWLEDGEMENT

We wholeheartedly dedicate this research to all who have been part of our journey towards the completion of this work.

First of all, to our beloved parents, thank you for the sleepless nights you worried with us, and for loving us through every tired moment.

We also dedicate this to our adviser and teachers, you didn't just teach. You listened, challenged us kindly, and pushed us to think deeper.

To our classmates, friends, and fellow researchers, who joined us in discussions, exchanged ideas, and provided emotional support, and shared laughter amid serious tasks.

And above all, to the Almighty God, who has been our guide and strength through all trials. To Him, we offer our eternal gratitude and praise.

This research is also dedicated to the generations of students to come. May this serve as a contribution to a broader understanding.

The authors



DEDICATION

We couldn't have finished this research without the people who stood by us. Thank you...

To the Almighty God, To God – for the strength to keep going when we wanted to stop, and for the wisdom we didn't know we had.

To our adviser, Ms. Rachelle Ann Ignacio, for your time, patience, and honest feedback. You didn't just guide us—you genuinely cared.

To our parents and family, for your understanding, for the quiet sacrifices, the moral support, and the financial help.

Our heartfelt thanks go to our classmates and friends, for walking with us, cheering us on, and reminding us to breathe when things got heavy.

And to each member of our group, thank you for the teamwork, understanding, and full dedication given to accomplish this research. This is ours.

Once again, our sincere gratitude to all of you.

The authors



TABLE OF CONTENTS

	Page
TITLE PAGE	i
APPROVAL SHEET	ii
ABSTRACT	iii
ACKNOWLEDGEMENT	iv
DEDICATION	v
TABLE OF CONTENTS	vi
LIST OF TABLES	viii
LIST OF FIGURES	x
LIST OF APPENDICES	x

Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction	1
Theoretical Framework	4
Conceptual Framework	5
Statement of the Problem	6
Null Hypothesis	7



Scope and Delimitation of the Study	8
Significance of the Study	8
Definition of Terms	10

Chapter 2

RESEARCH METHODOLOGY

Research Design	13
Locale of the Study	14
Respondents	15
Research Instrument	16
Validation of the Instrument	17
Data Gathering Procedure	18
Statistical Treatment of Data	19
Interpretation of Scores	21
Ethical Considerations	23

Chapter 3

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA	26
--	-----------



Chapter 4

SUMMARY, FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

Summary of Findings	48
Conclusions	49
Recommendations	50

BIBLIOGRAPHY	54
---------------------	----

APPENDICES	57
-------------------	----

Appendix A Letter of permission to conduct the study	58
--	----

Appendix B Questionnaire	68
--------------------------	----

Table of Specification	76
------------------------	----

Appendix C Normality Test	77
---------------------------	----

Appendix D Interpretation of Scores	79
-------------------------------------	----

Appendix E Certificate of Statistician	81
--	----

Appendix F Compliance Report	82
------------------------------	----

Appendix G RERU Protocol Approval	88
-----------------------------------	----

CURRICULUM VITAE	90
-------------------------	----

LIST OF TABLES

Table	Page
--------------	-------------



1	Interpretation of Scores: Color, Shape Recognition, and Sequencing	21
2	Interpretation of Scores: Number Recognition and Counting	22
3	Interpretation of Scores: Readiness for Basic Operations	23
4	Profile of Respondents: Age and Sex	26
5	Mathematical Literacy During Pretest	28
6	Mathematical Literacy During Posttest	30
7	Distribution of Kindergarten Pupils according to Skill level in terms of Color, Shape Recognition, And Sequencing	33
8	Distribution of Kindergarten Pupils according to Skill level in terms of Number Recognition and Counting	35
9	Distribution of Kindergarten Pupils according to Skill Level in terms of Readiness for basic Operation	37
10	Distribution of Kindergarten Pupils according to Skill Level in terms of Mathematical Literacy	39
11	Significant Levels of Overall mathematical Literacy in Terms of Age	41



12	Significant Levels of Overall mathematical Literacy in Terms of Sex	43
13	Significant Levels of Overall mathematical Literacy During Pretest and Posttest	45

LIST OF FIGURES

Figure	Page
1 Conceptual Paradigm of the Study	5
2 The Location Map of the Locale of the Study	15

LIST OF APPENDICES

Appendix	Page
A Letter of Permission to Conduct the Study	57
B Questionnaire: The Instrument	68
Table of Specification	76
C Normality Test	77
D Interpretation of Scores	79



Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

The formative years of childhood served as the cornerstone for a lifetime of learning and development. In this developmental window, children learned not only the basic operations but even more, acquired the foundation for reasoning, and understood logic. Research showed that early mathematical literacy emerged not as the precursor to formal arithmetic but as the necessity of cognitive architecture (National Research Council, 2018). Moreover, the National Association for the Education of Young Children (NAEYC, 2020) placed mathematics as one of the most important areas in the best quality early childhood programs, declaring that very young children had an extraordinary, unlearned ability for mathematical thinking which, if allowed to grow, became an important factor for academic success in the future.

The international landscape had been supportive of early numeracy skills as being the most important skill of all and the most lasting one. In the same light, the classic longitudinal research of Watts et al. (2018) discovered that the ability in preschool mathematics was one of the strongest indicators of the mathematics achievement in high school, directly indicating the competence of the early set. Furthermore, Purpura et al. (2017) laid out the scenario where early numeracy skills stood out as a stronger and more direct predictor of later mathematics



achievement when compared to early literacy, attention, or socioemotional skills. Thus, the research findings concluded that even the very first signs of deficiency in math at the kindergarten level should be taken very seriously; they could cause a chain of learning issues starting from the minor ones, and this was often referred to as the "Matthew Effect" in mathematics, which happened when the latter was too late and the initial gap became wider over time (Nguyen et al., 2016).

In the Philippine setting, the passing of the Kindergarten Education Act, also known as Republic Act 10157, along with the setup of the Mother Tongue-Based Multilingual Education (MTB-MLE) program, showcased the government's full commitment to the development of basic competencies. Despite this, the commitment was straining under the burden of the dismal performance of the country's learners in international assessments. The 2022 Programme for International Student Assessment (PISA) revealed a continued challenge, with 84% of Filipino 15-year-old students scoring below the minimum proficiency level in mathematics (OECD, 2023). Though these tests basically targeted older students, they reflected a predicament of the whole education system that probably had roots in the early years. The issue could have been further aggravated by a possible gap in the way subjects were taught; Lim's (2022) investigation of the Philippine early childhood education found that teachers acknowledged the importance of play, yet they were often not well equipped with



the required pedagogical content knowledge to create and carry out effective play-based activities that would promote mathematical thinking.

The merge of the worldwide findings and the local difficulties indicated a critical need to “count the uncounted” — to conduct systematic evaluations and comprehend the mathematical literacy of the learners who were just about to set foot on their academic journey. It would be very hard to know exactly what the points of intervention and support were in early math literacy without a clear and thorough data-driven understanding of the specific strengths and weaknesses; hence, the interventions at the kindergarten level would be at risk of being generic, misaligned, or ineffective.

This research was situated within the context of the aforementioned issue, seeking to examine the foundational numeracy skills of young learners, recognizing their crucial role in shaping later academic achievement and addressing the broader challenges reflected in national and international assessments. The investigation was based on the premise that number recognition, counting, and readiness for basic operations were the key domains of mathematical literacy and therefore would provide an empirical snapshot of the mathematical landscape for young learners in a specific local context.

This research aimed to assess kindergarteners' fundamental mathematical literacy levels at New Cabalan Elementary School, in response to the systemic mathematical deficiencies indicated in national assessment data. This research



started with solid, reliable data specifically looking at children's age and sex to better understand how to support their mathematical growth. By doing so, it aimed to guide evidence-based teaching practices that met young learners where they were, helping build a stronger and more equitable foundation in math.

Theoretical Framework

Here is the text revised into the past tense where necessary, while keeping the original meaning and structure intact:

The Learning Trajectories Approach proposed by Clements and Sarama back in 2021 was used as an anchor for the study. The approach consisted of three main pillars: the goals, the developmental progression, and the instructional materials.

The first component, "goals", referred to the areas to be learned, involving the creation of specific objectives for gauging purposes later on, which specifically targeted endogenous development factors later on, establishing the target mathematical competence that was expected of the kindergarten learners.

The second component, the developmental progression of students, emphasized the need to observe children during learning and to understand the natural trajectory of their mathematical learning, and posited that learners could advance through different learning levels.

The last component was the instructional materials, which stated that instructions were made to clear the gap between the current abilities of a learner

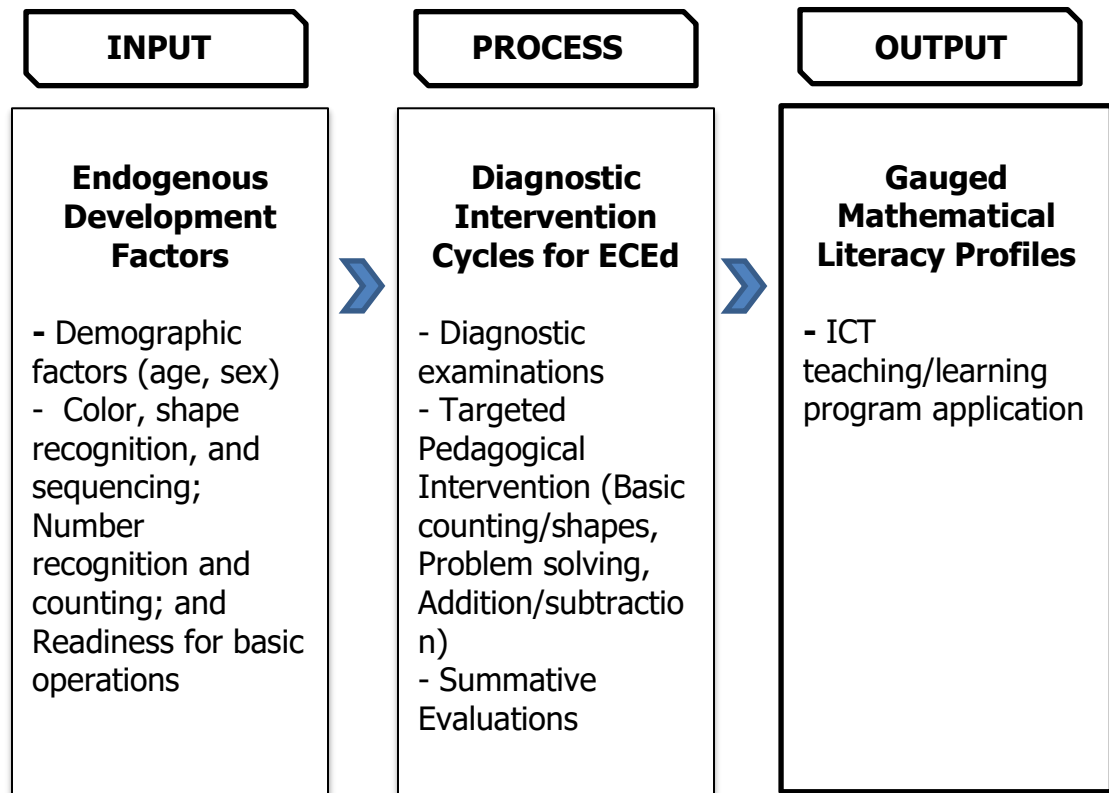


and their potential, mainly through the creation of an assessment tool to gauge their levels of mathematical literacy and the intervention implementations needed to scaffold their learning.

Conceptual Framework

Figure 1

Conceptual Framework of the Study



As seen in Figure 1, this level consists of three (3) frames. The first frame indicated the input. In exploring the mathematical literacy levels among kindergarteners at New Cabalan Elementary School, child-specific factors such as



age, sex, number recognition and counting, readiness for basic operations, shape-recognition capabilities, and overall mathematical literacy will play a crucial role.

The second frame represented the process. To explore the mathematical literacy levels among kindergarteners at New Cabalan Elementary School, the researchers administered a test, implemented in the context of kindergarten mathematics instruction. The content focused on essential areas such as basic counting, shape recognition, and foundational problem-solving strategies. This comprehensive approach aimed to provide a clear picture of each child's mathematical abilities and inform future instructional strategies.

The third frame represented the output. The results of the study offered valuable fresh insights on the mathematical competency of the kindergarteners at New Cabalan Elementary School. This study provides a crucial understanding of students' present competence levels, guiding future instructional strategies to support their mathematical development.

Statement of the Problem

This study aims to gauge the level of mathematical literacy of kindergarten pupils in New Cabalan Elementary School.

Specifically, it aims to answer the following questions:

1. How may the number and percentage of the respondents be described in terms of:



- 1.1. Age, and
- 1.2. Sex?
2. What is the level of mathematical literacy of male and female kindergarten pupils before and after intervention in terms of:
 - 2.1. Color, shape recognition, and sequencing;
 - 2.2. Number recognition and counting; and
 - 2.3. Readiness for basic operations
3. Is there a significant difference between the levels of overall mathematical literacy in terms of age?
4. Is there a significant difference between the level of overall mathematical literacy in terms of sex?
5. Is there a significant difference between the level overall mathematical literacy before and after the intervention?
6. What new teaching model, program, paradigm, or instructional framework can be proposed based on the study's findings to enhance kindergarten mathematical literacy?

Hypothesis

H_0 : There is no significant difference in the mathematical literacy levels among kindergarteners in terms of age and sex in New Cabalan Elementary School.



Scope and Delimitations of the Study

This study is limited to assess the mathematical literacy levels among kindergarteners at New Cabalan Elementary School during the School Year 2025-2026.

It is delimited to one public elementary school, two sections of kindergarten students of New Cabalan Elementary School as respondents of this study. This study has taken into consideration the age and sex of the students.

The respondents were selected purposively, a total of 42 students across two kindergarten class sections handled by the same teacher from New Cabalan Elementary School, Olongapo City A.Y. 2025-2026. The researchers used test administration instrument as the main instrument in gathering data and information of the respondents. The implementation itself for attaining the desired number of respondents and the overall gathering and collation of data take effect for two weeks.

Significance of the Study

The findings of the study were found to be beneficial to the following:

Kindergarten learners. In this study, the kindergarten learners were assessed of their mathematical literacy levels. This study can help identify the



learning gaps of the learners which can help to provide appropriate interventions in supporting their needs and interest in mathematical literacy.

Teachers. This study may provide ideas to teachers concerning the mathematical literacy of their learners. This study can provide insights to the current levels of mathematical literacy of the students to be able to design more developmentally appropriate lessons to enhance mathematical skills of their teachers and the learners.

Parents/guardians. This study can provide awareness to the parents or guardians to provide better support at home. This study will help them to make better decisions and plans to help their children in their mathematics skills.

School Administrators. This study can serve as a basis for planning and allocating resources that will be beneficial in providing essential materials for the learners. Thus, it may provide further knowledge for improvement of teaching styles.

Department of Education. This study may be beneficial in addressing the problem of mathematical literacy of Filipinos by starting in early levels. Through the help of this study, the results may help the Department of Education in enhancing the early mathematics curriculum and strengthen the foundation of mathematical skills in the early grades.



Future Researchers. This study will provide information to all early childhood researchers that may be find helpful to their research associated or similar to their research topic. Additionally, this study may be found beneficial for other researchers in exploring deeper issues related to this matter.

Definition of Terms

For the sake of clarity and understanding, some terms in this study have been defined in the study as follows:

Counting Skills. Refers to the ability to recite numbers in sequence and establish one-to-one correspondence between numbers and objects (Clements & Sarama, 2021). In the study, this is measured through the kindergarteners' ability to count numbers from 1 to 20 and match them with corresponding quantities during the assessment.

Early Numeracy. Refers to the foundational understanding of numbers, quantities, and basic mathematical relationships developed in early childhood (Purpura, Reid, Eiland, & Baroody, 2015). In the study, it is determined by the kindergarteners' performance in number recognition, counting, and readiness for basic operations.

Endogenous Development. It is a developmental approach that emerges from within a community, area, or territory, depending mainly on local resources,



knowledge, culture, institutions, and participants rather than external sources, according to Adam, Viana, and Valle (2025). In the study, it is the child's development from the start in terms of what they already know about the topic or subject and their potential trajectories.

Diagnostic Intervention Cycle. It is the iterative process utilized in facilitation and organizational development, when a team or practitioner observes, makes sense of what they see, determines what to do, and then acts (Chapman, 2017). In the study, it is a full cycle that teachers use in order to observe the children, orient on the problems they see, decide on the intervention methodology they will do on the spot, and conduct said intervention to refine student mathematical understanding.

Mathematical Literacy. Refers to the ability to understand and apply mathematical concepts to interpret and solve everyday problems (OECD, 2023). In the study, it is measured through the kindergarteners' performance in tasks related to number recognition, counting, and basic mathematical operations.

Mathematical Literacy Level. Refers to the degree or stage of an individual's understanding and application of mathematical concepts (OECD, 2019). In the study, it is determined based on the respondents' scores and categorized levels in the pretest and posttest conducted by the researchers.

Number Recognition. Refers to the ability to identify and name numerical symbols accurately (Clements & Sarama, 2021). In the study, it pertains to the skill



of correctly identifying and naming numbers from 0 to 20 as evaluated through test items.

Readiness for Basic Operations. Refers to a child's preparedness to perform simple arithmetic processes such as addition and subtraction (Baroody, 2017). In the study, it is assessed through the kindergarteners' ability to demonstrate emerging understanding of basic addition and subtraction during the test.



Chapter 2

RESEARCH METHODOLOGY

This chapter outlines the systematic procedures and strategies that will be employed to conduct the study. It includes a detailed description of the research design, the participants and sampling procedure, the data collection instrument, the data gathering procedure, and the statistical treatment of data. This framework is designed to ensure the study is conducted in a valid, reliable, and ethical manner.

Research Design

This study will employ a descriptive-comparative research design. The descriptive aspect of the design is concerned with assessing and describing the level of mathematical literacy of the kindergarten pupils in terms of number recognition, counting, and readiness for basic operations. This involves collecting data to provide a detailed snapshot of the current state of these variables (Fraenkel et al., 2019). The comparative aspect will be used to determine if significant differences exist in the overall mathematical literacy levels when the participants are grouped by age and sex.

The comparative dimension of this design is activated when analyzing the collected data to determine if significant differences exist in the overall mathematical literacy levels based on the independent variables of age and sex. As articulated by Fraenkel et al. (2019), this approach is methodologically sound



as it facilitates both the quantitative description of group characteristics and the statistical comparison of means to test prespecified hypotheses regarding the population.

Locale of the Study

The study was conducted in New Cabalan Elementary School, located at 28 Rizal Street, Olongapo City, 2200 Zambales. Specifically, the research focused on the kindergarten department where young learners are beginning to develop their foundational skills in mathematics. The study aimed to explore and assess the mathematical literacy levels among kindergarten pupils.

New Cabalan Elementary School is a public institution under the Department of Education that provides quality basic education to children in the community of Olongapo City. The school offers complete elementary education, from kindergarten to grade six, and is known for its dedicated teachers and supportive learning environment. It continues to uphold its mission of nurturing well-rounded learners equipped with the essential knowledge and skills for lifelong learning.

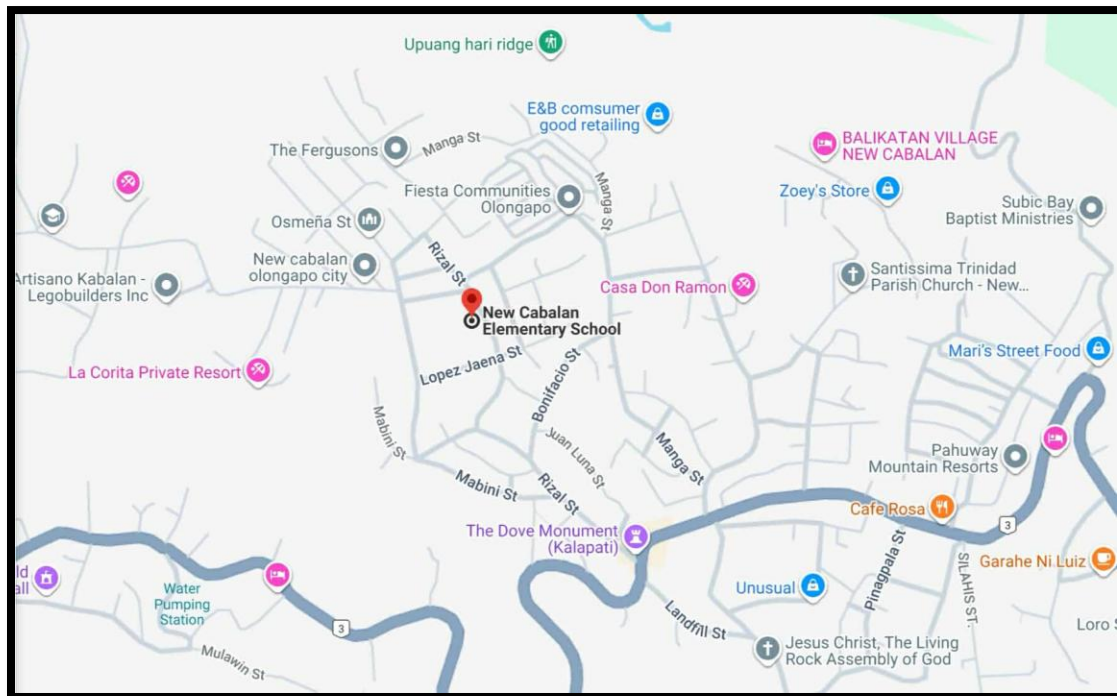
The researchers selected New Cabalan Elementary School due to its distance from the main city proper. According to the teachers, the school is rarely chosen for research, providing a less-studied setting. More remote schools, such as Iram Elementary School and Amelia Heights Elementary School, were not chosen for practical reasons, with the researchers prioritizing time management and logistical ease for the study's implementation.



The researchers would also like to conduct a study involving the NCES, specifically Math for Kindergarten classes, as there has been no other research undertaken on this topic in the city of Olongapo. Specifically targeting the topic of mathematics, it will be noted that the Philippines has a low ranking on the topic of PISA rankings. Meanwhile, since no one has conducted research on kindergarten in Olongapo in the field of Mathematics, we aim to start this to aid future researchers and benefit the children.

Figure 2

The Location Map of the Locale of the Study



Respondents

The respondents of this study were the kindergarten pupils enrolled in New Cabalan Elementary School in Olongapo City during the School Year 2025 to 2026.



The researchers employed purposive sampling as it was considered the most appropriate method for the study. This sampling technique allowed the selection of participants who possessed specific characteristics relevant to the research, making it practical and suitable given the available time, resources, and accessibility of the respondents (Creswell, 2014).

A total of forty-two kindergarten pupils aged five to six years old participated in the study. Twenty-one pupils came from Section Mickey Mouse and twenty-one from Section Donald Duck. These pupils were carefully chosen as they represented the group of learners whose early mathematical literacy was the primary focus of the study, mostly due to their class standings. Prior to their participation, the researchers secured consent from the parents or guardians to ensure voluntary and well-informed involvement.

The respondents were involved in the study through the administration of a pre-test and a post-test. The pre-test was conducted to assess their initial level of mathematical literacy, while the post-test was administered afterward to measure any improvement following the implementation of the learning activities. The gathered results provided valuable data in determining the effectiveness of the designed activities in enhancing the pupils' mathematical literacy skills.

Research Instrument

The main instrument in data gathering is test administration consisting of 20 items. The type of test administered used in the study measured the level of



mathematical literacy of two sections of kindergarten in New Cabalan Elementary School. It allows to help assess the overall mathematical literacy of the kindergarteners.

The researchers administered a consent form to the parents to inform them about their children's involvement as the respondents. Once the consent was secured, the researchers utilized a paper-pencil test which was analyzed to measure their level of mathematical literacy.

The 20-items pre-test and post-test assessment contains of the same content are made with a Table of Specifications (TOS) derived from the Matatag MELCs ensuring that the items are appropriate for the age and development of the learners.

Validation of the Instrument

The instrument developed by the researchers, which is a test administered to assess the learner performance, underwent a validation process to ensure its appropriateness and alignment with early childhood principles. The validation was conducted with the guidance and of the research adviser, and through the formal evaluation of a panel of experts.

The validators consist of two Doctors of Education and one is a Doctor of Philosophy specializing in early childhood. The panel of experts were selected for



having an expertise and experience in early childhood education, given that, they assessed the 20-item pretest and posttest for age-appropriateness, language clarity and relevance for the numeracy of kindergarten ensuring that there will be no confusion for the learners. The feedback that was given were revised in the instrument accordingly.

Data Gathering Procedure

In gathering data, a formal request letter will first be sent to the Office of the School Principal of New Cabalan Elementary School. This letter will ask for permission to conduct the study on school grounds. It will explain the study's goals, the participants involved, and the general procedures to be followed. Data collection will begin only after the school administration approves the request.

Subsequent to this, parental consent forms shall be distributed to parents/guardians of prospective participant respondents. These forms carefully describe study purposes, procedures involved in carrying out the study, guarantees of confidentiality, and voluntary participant schemes. Parents/guardians shall be made aware that there would be no untoward effects on withdrawal for their wards prior to participation in the study. Only participant respondents with parental consent forms would be used in this study.



Third, the kindergarten teacher participating in the study will receive an orientation from the researchers. To ensure appropriate coordination and cooperation, the goals of the study, the methods for assessment, and the roles of the teachers will all be explained during this orientation. The purpose of this step is to make sure that the assessment procedure runs smoothly and that the students are comfortable and well-prepared for the task.

In order to provide a comfortable setting for the students, the next step, which involves the actual assessment, will then take place in a familiar and quiet part of the school. To maintain focus and engagement, each child will be evaluated separately during a single session that lasts roughly 15 to 20 minutes. The researchers will make sure that the processes are non-threatening and kid-friendly.

Lastly, a methodical compilation, encoding, and organization of all the data collected from the assessment sheets will take place. After that, the prepared data will undergo the necessary statistical analysis to meet the study's goals. Every record will be kept private and used only for research and academic purposes.

Statistical Treatment of Data

This study employs quantitative data analysis through the use of a test administration as a major instrument for the analysis of the mathematical literacy among kindergarten students within New Cabalan Elementary School. Based on



the objectives identified in this study, the statistical tools to be used will be descriptive statistics and inferential statistics.

Frequency and Percentage

Frequency and percentage were employed as basic descriptive statistics to summarize the distribution of kindergarten pupils' demographic profiles, specifically age and sex. Frequency refers to the simple count of cases within each category, while percentage expresses that count relative to the total sample allowing for comparisons across groups.

Mean and Median

The mean is the average and the median is the middle score when data are arranged from lowest to highest. Because most scores were not normally spread out, the median was used to decide if the learners were "Skilled" or "Highly Skilled."

Normality Test

The Shapiro–Wilk test was then employed to assess whether the pretest and posttest scores across all mathematical literacy domains are in a normal distribution, as this assumption is necessary for parametric tests such as t-tests or ANOVA.



Wilcoxon Signed-rank Test

Then, Wilcoxon signed-rank test which is a non-parametric alternative to the paired t-test that compares median differences between two related samples was then employed to answer research question 5, specifically, whether there was a significant improvement in overall mathematical literacy from pretest to posttest among the same group of kindergarten pupils.

Mann-Whitney U Test

Mann-Whitney U test is a non-parametric alternative to the independent t-test that compares the ranks of scores rather than means which was used to determine whether overall mathematical literacy differed significantly between two independent groups based on age and sex.

Rank-biserial Correlation

The rank-biserial correlation was calculated as a measure of effect size for Mann-Whitney U and Wilcoxon tests to assess the practical significance of the findings beyond the reported p-values. According to (Peres, 2025), values of $|0.10|$ indicate a small effect, $|0.30|$ indicate a medium effect, and $|0.50|$ indicate a large effect.

INTERPRETATION OF SCORES:

Table 1



Color, Shape Recognition, and Sequencing

Raw Score Range	% Range	Skill level	Interpretation
0 – 1	60 – 66.67	Not Yet Skilled	Cannot recognize colors/shapes; no sequencing skill
2 – 3	70 – 74	Low or Limited skill	Low – Limited recognition; sequencing needs assistance
4	75 – 80	Moderately Skilled	Basic recognition and sequencing achieved
5	85 – 90	Skilled	Good recognition and sequencing skills
6	95 – 100	Highly Skilled	Mastery of recognition and sequencing

Table 2

Number Recognition and Counting

Raw Score Range	% Range	Skill Level	Interpretation
0 – 2	60 – 66.67	Not Skilled	Cannot recognize numbers or count reliably
3 – 6	70 – 74	Limited Skilled	Recognizes some numbers and counts with assistance
7	75 – 80	Moderately Skilled	Recognizes numbers and counts with basic accuracy
8 – 9	83 – 88	Skilled	Accurately recognizes numbers and counts with minimal errors



10 – 12	90 – 100	Highly Skilled	Demonstrates mastery in number recognition and counting
---------	----------	----------------	---

Table 3

Readiness for Basic Operations

Raw Score Range	% Range	Skill Level	Interpretation
0	60	Not Yet Skilled	No understanding of operations
1	75 – 80	Skilled	Emerging readiness for operations
2	95 – 100	Highly Skilled	Ready for basic operations independently

Ethical Considerations

The study will adhere to established ethical principles in conducting research involving young children. The framework is guided by the Belmont Report's principles of respect for persons, beneficence, and justice, as well as the ethical guidelines for research with children outlined by the National Association for the Education of Young Children (NAEYC, 2020), and the Philippine Department of Education.

First, an informed consent is obtained from the parents or legal guardians of all prospective child participants. The consent form clearly explained the study's



purpose and direction, alongside its future benefits and the duration. It also explained the voluntary nature of participation of the students and their rights to be able to withdraw from the study at any given time with no penalties, which includes the clauses of the Data Privacy Act of 2012. Alongside these, formal written approvals are secured from the School Principal of New Cabalan Elementary School and the relevant DepEd authorities.

Next, assessments that are dish out were designed to be low-risk, non-invasive, age-appropriate, while also providing benefit to teachers with data to use for better instructional support, while ensuring that all results aren't used to discriminate.

Third, participants are selected purposively with no discrimination used regarding their backgrounds, performance, and abilities. All participants are ensured and treated fairly, undergoing the same treatment as the others, in order to ensure fairness in collection of data.

Data collected from the study are all anonymized, where participant names are replaced with unique identification codes in all records to prevent potential data leaks. Alongside that, physical assessment sheets and collected consent forms are all placed in a secure location, and all digital files are placed in password-protected drives, while findings are reported in aggregate form with no individual



child performance being visible, and there are no cooked, falsified, or fabricated data.

Lastly, a general summary of the findings are shared with the school administration and the kindergarten teachers to support their instructional planning, and all physical & digital data involving sensitive information will be destroyed and deleted.



Chapter 3

PRESENTATION, ANALYSIS AND INTERPRETATION OF DATA

This chapter presents the findings, analysis, and interpretation of the data gathered from the kindergarten pupils of New Cabalan Elementary School. It includes the profile of the respondents, the level of mathematical literacy before and after the intervention, and the comparative analysis of the variables. The presentation follows the sequence of the statement of the problem. Specifically, it aims to answer the following questions:

Table 4

Profile of the Kindergarten Pupils in terms of Age and Sex

Profile	Frequency	Percentage
Age		
5	22	52.4
6	20	47.6
Sex		
Male	20	47.6
Female	22	52.4
Total	42	100

Table 4 presents the frequency and percentage distribution of the respondents when grouped according to their age and sex. In terms of age, 22 or 52.4% of the kindergarten pupils are aged 5 years old, while 20 pupils or 47.6%



are aged 6 years old. In terms of sex, 22 or 52.4% of the respondents are female, and 20 pupils or 47.6% are male.

The age distribution of kindergarten pupils in this study reflects the typical age range for kindergarten enrollment in the Philippine educational system. Under the Department of Education's Kindergarten Education Act (Republic Act 10157), the standard age for kindergarten is five years old, with six-year-olds often representing those who may have started later or been retained. The nearly equal distribution between five and six-year-olds (52.4% and 47.6%, respectively) provides a balanced representation of both age groups, allowing for meaningful comparisons in mathematical literacy development.

The sex distribution shows a slightly higher number of female respondents (52.4%) compared to male respondents (47.6%). This slight female predominance is consistent with typical enrollment patterns in Philippine public schools, where gender distribution tends to be relatively balanced. This distribution is important as it allows for an adequate examination of potential gender-based differences in mathematical literacy among kindergarten learners. Research by Chang, Sandhofer, and Brown (2020) suggests that early number exposure and mathematical development may vary by gender due to socialization practices, making this balanced representation valuable for the study's comparative analyses.



Table 5

Mathematical Literacy During Pretest

Mathematical Literacy	Mean	Median	Interpretation (Median)
Color and Shape Recognition, and Sequencing	4.60	5.00	Skilled
Number Recognition and Counting	8.40	9.00	Skilled
Readiness for Basic Operation	1.79	2.00	Highly Skilled
PRETEST OVERALL	14.79	16.00	Highly Skilled

Table 5 illustrates the baseline mathematical literacy of the participants across three core domains and their overall performance prior to the intervention. In the area of Color and Shape Recognition, and Sequencing, the participants achieved a mean score of 4.60 and a median of 5.00. According to the established assessment rubric, a median score of 5.00 corresponds to a "Skilled" level, indicating that the learners already possessed a solid ability to recognize colors and shapes and could perform sequencing tasks with minimal guidance before the study commenced.

The results for Number Recognition and Counting showed a mean of 8.40 and a median of 9.00. This median value falls within the "Skilled" range (8–9),



suggesting that the participants were capable of accurately recognizing numbers and counting with high reliability. This finding aligns with the developmental progression outlined by Clements and Sarama (2021), who posited that children at the kindergarten level typically develop counting competencies through exposure and practice. The median score of 9.00 indicates that more than half of the participants achieved near-perfect scores in this domain, reflecting a strong foundational understanding of numerical concepts.

The domain of Readiness for Basic Operations yielded the highest quantitative performance, with a mean of 1.79 and a median of 2.00. A median score of 2.00 is classified as "Highly Skilled," demonstrating that the participants were independently prepared for basic mathematical operations at the start of the intervention. This suggests that despite the participants being in the kindergarten level, they exhibited emerging readiness for addition and subtraction tasks. However, it is important to note that this domain had a maximum score of 2, which may have created a ceiling effect. The highly skewed distribution (Shapiro-Wilk $W = 0.424$, $p < .001$) indicates that the majority of participants scored the maximum points, leaving little room for growth. This finding warrants careful interpretation, as the assessment may have been too limited to capture the full range of readiness for basic operations.

The Pretest Overall mathematical literacy results presented a cumulative mean of 14.79 and a median of 16.00. While the mean score ($M=14.79$) sits within the



"Skilled" category (13–15), the median score (Mdn=16.00) officially classifies the group's performance as "Highly Skilled" (16–20). This statistical disparity, confirmed by the Shapiro-Wilk test for normality ($W = 0.836$, $p < .001$), indicates a negative skew in the data, implying that while a small number of lower scores pulled the arithmetic average down, the majority of the participants demonstrated mastery of the subject matter. Consequently, the interpretation based on the median confirms that the participants entered the intervention with an advanced foundational understanding and a high level of mathematical literacy.

This finding is supported by Watts et al. (2017), who stated that early mathematics interventions have positive effects on children's mathematical achievement and learning processes. Similarly, the improvement observed in this study suggests that appropriate instruction and intervention can strengthen foundational mathematical literacy among kindergarten learners.

Table 6

Mathematical Literacy During Posttest

Mathematical Literacy	Mean	Median	Interpretation (Median)
Color and Shape Recognition, and Sequencing	4.93	5.00	Skilled
Number Recognition and Counting	9.24	10.00	Highly Skilled



Readiness for Basic Operation	1.86	2.00	Highly Skilled
POSTTEST OVERALL	16.02	16.00	Highly Skilled

The post-intervention assessment of mathematical literacy, as presented in Table 6, reflects the participants' progress following the implementation of the program. In the domain of Color and Shape Recognition, and Sequencing, the participants achieved a mean (M) of 4.93 and a median of 5.00. According to the established rubric, a median of 5.00 remains classified at the "Skilled" level. While the qualitative category did not shift, the increase in the mean from the pretest (4.60) to the posttest (4.93) suggests a positive trend toward higher consistency and mastery within this skill set. The improvement indicates that the intervention reinforced these foundational skills, allowing learners to approach ceiling-level performance in this domain.

Significant growth was observed in Number Recognition and Counting, where the participants reached a mean of 9.24 and a median of 10.00. This median score represents a qualitative advancement from the pretest "Skilled" level to the "Highly Skilled" category (10–12). This shift indicates that, following the intervention, the majority of learner's demonstrated mastery of number identification and counting, transitioning from minimal errors to a professional level of accuracy and independence. This improvement is particularly noteworthy given that the pretest median was already high (9.00). The ability to achieve a perfect median score of



10.00 suggests that the intervention successfully addressed any remaining gaps in number recognition and counting skills. The Shapiro-Wilk test for the Number Recognition and Counting posttest ($W = 0.925$, $p = .009$) indicates that while the distribution remained non-normal, it approached normality more closely than the pretest, suggesting more uniform performance among participants.

In the area of Readiness for Basic Operations, the participants maintained a high level of performance with a mean of 1.86 and a median of 2.00. The median score of 2.00 confirms that the participants remained "Highly Skilled," reinforcing their independent readiness to engage with operational mathematical tasks. However, as noted in the pretest analysis, the ceiling effect in this domain (maximum score of 2) limits the ability to detect meaningful improvement. The highly skewed distribution persisted in the posttest (Shapiro-Wilk $W = 0.417$, $p < .001$), with 36 out of 42 participants scoring the maximum 2 points. This pattern suggests that the assessment of readiness for basic operations may need to be expanded to include more complex tasks to better differentiate skill levels among learners who have already mastered the foundational skills.

The Posttest Overall mathematical literacy results showed a cumulative mean of 16.02 and a median of 16.00. Both measures now align within the "Highly Skilled" range (16–20). Notably, the mean increased from 14.79 in the pretest to 16.02 in the posttest, which effectively eliminated the previous negative skew, as evidenced by the Shapiro-Wilk test approaching normality ($W = 0.947$, $p = .053$).



This indicates that the intervention successfully moved the lower-performing students into the higher competency brackets, resulting in a more uniform group performance. The improvement in overall mathematical literacy by 1.23 points represents a meaningful gain, particularly considering the already high baseline performance. This finding aligns with the research of Watts et al. (2018), which demonstrated that early mathematics interventions can have significant positive impacts on mathematical achievement, even for children who already demonstrate foundational skills.

Table 7

Distribution of Kindergarten Pupils according to Skill Level in terms of Color, Shape Recognition, And Sequencing

Skill Level	Interpretation	Score Range	Pre-test (f)	Post Test (f)
Not Yet Skilled	Cannot recognize colors/shapes; no sequencing skill	0 – 1	0	0
Low or Limited skill	Low – Limited recognition; sequencing needs assistance	2 – 3	4	1
Moderately Skilled	Basic recognition and sequencing achieved	4	14	13
Skilled	Good recognition and sequencing skills	5	18	16



Highly Skilled	Mastery of recognition and sequencing	6	6	12
Total			42	42

Table 7 presents the distribution of kindergarten pupils according to their skill levels in color and shape recognition and sequencing. Prior to the intervention, the majority of participants were classified as "Skilled" (18 pupils, 42.9%) and "Moderately Skilled" (14 pupils, 33.3%). A notable number of pupils (6 or 14.3%) were classified as "Highly Skilled" even before the intervention, while 4 pupils (9.5%) fell into the "Low or Limited skill" category. No pupils were classified as "Not Yet Skilled" in the pretest.

Following the intervention, the number of pupils in the "Highly Skilled" category doubled from 6 to 12 (28.6%), representing a substantial increase of 100% in the mastery level. Meanwhile, the number of pupils in the "Low or Limited skill" category decreased from 4 to 1 (2.4%), indicating that the intervention successfully addressed the learning gaps of most pupils who initially struggled with these concepts. The "Skilled" category decreased slightly from 18 to 16, which may be attributed to pupils moving into the "Highly Skilled" category. Similarly, the "Moderately Skilled" category decreased from 14 to 13, reflecting the overall upward shift in skill distribution.

This pattern of improvement demonstrates the effectiveness of the intervention in enhancing color and shape recognition and sequencing skills. The doubling of pupils achieving mastery level is particularly significant, as it suggests that the



instructional activities provided sufficient scaffolding for learners to progress from proficient to advanced levels. This finding is consistent with Vygotsky's (1978) theory of the Zone of Proximal Development, wherein learners are able to achieve higher levels of competency through guided instruction and appropriate scaffolding.

Table 8

Distribution of Kindergarten Pupils according to Skill Level in terms of Number Recognition and Counting

Skill Level	Interpretation	Score Range	Pre-test (f)	Post Test (f)
Not Skilled	Cannot recognize numbers or count reliably	0 – 2	2	0
Low or Limited Skilled	Recognizes some numbers and counts with assistance	3 – 6	6	6
Moderately Skilled	Recognizes numbers and counts with basic accuracy	7	1	1
Skilled	Accurately recognizes numbers and counts with minimal errors	8 – 9	17	13
Highly Skilled	Demonstrates mastery in number recognition and counting	10 – 12	16	22
Total			42	42

Table 8 presents the distribution of kindergarten pupils according to their skill levels in number recognition and counting. Prior to the intervention, the largest



groups of pupils were classified as "Skilled" (17 pupils, 40.5%) and "Highly Skilled" (16 pupils, 38.1%). However, 6 pupils (14.3%) were classified as "Low or Limited Skilled," and 2 pupils (4.8%) were classified as "Not Skilled." This distribution indicates that while a substantial proportion of participants demonstrated strong number sense, there was a notable group of learners who required additional support in this foundational skill area.

Following the intervention, the number of pupils in the "Highly Skilled" category increased from 16 to 22 (52.4%), representing a 37.5% increase in mastery-level performance. Importantly, the 2 pupils who were initially classified as "Not Skilled" moved out of this category entirely, demonstrating that the intervention successfully addressed even the most foundational learning gaps. The "Skilled" category decreased from 17 to 13, which may reflect the upward movement of pupils into the "Highly Skilled" category. The "Low or Limited Skilled" category remained stable at 6 pupils (14.3%), suggesting that while the intervention was effective for most learners, a subset of pupils may require more intensive or sustained support to develop proficiency in number recognition and counting.

The significant increase in the number of pupils achieving mastery level (from 38.1% to 52.4%) is particularly noteworthy given that number recognition and counting form the foundation for later mathematical achievement. According to Purpura and Lonigan (2016), early numeracy skills, particularly number recognition and counting, are critical predictors of later mathematics achievement. The



improvement observed in this domain aligns with the findings of Nguyen et al. (2016), who emphasized that early intervention in numeracy skills can prevent the "Matthew Effect" in mathematics, where initial gaps widen over time. The fact that the intervention successfully moved the two "Not Skilled" pupils out of that category suggests that targeted instructional support can effectively address early numeracy deficits.

Table 9

Distribution of Kindergarten Pupils according to Skill Level in terms of Readiness for Basic Operation

Skill Level	Interpretation	Score Range	Pre-test (f)	Post Test (f)
Not Yet Skilled	No understanding of operations	0	3	0
Skilled	Emerging readiness for operations	1	3	6
Highly Skilled	Ready for basic operations independently	2	36	36
Total			42	42

Table 9 presents the distribution of kindergarten pupils according to their skill levels in readiness for basic operations. Prior to the intervention, an overwhelming majority of pupils (36 pupils, 85.7%) were classified as "Highly Skilled," indicating independent readiness for basic operations. Three pupils (7.1%) were classified as "Skilled," showing emerging readiness, while three pupils (7.1%) were classified as "Not Yet Skilled," demonstrating no understanding of operations.



Following the intervention, the number of pupils in the "Not Yet Skilled" category decreased from 3 to 0, indicating that the intervention successfully helped these pupils develop at least emerging understanding of basic operations. The "Skilled" category increased from 3 to 6 (14.3%), while the "Highly Skilled" category remained stable at 36 pupils (85.7%). This pattern suggests that the intervention was particularly effective for the lowest-performing pupils, moving them into the emerging readiness category.

The exceptionally high proportion of pupils in the "Highly Skilled" category both before and after the intervention warrants careful interpretation. As noted in the analysis of mean scores, this domain had a maximum possible score of 2, which created a ceiling effect that limited the ability to detect growth among the majority of participants. The assessment items for readiness for basic operations may have been too simplistic for this group of learners, many of whom already demonstrated mastery of these concepts. According to Baroody (2017), readiness for basic operations encompasses a range of skills beyond simple addition and subtraction, including understanding part-whole relationships and counting-on strategies. Future assessments may benefit from including more complex items to better differentiate skill levels among learners who have already mastered foundational operational concepts.



Table 10

Distribution of Kindergarten Pupils according to Skill Level in terms of Overall Mathematical Literacy

Skill Level	Interpretation	Score Range	Pre-test (f)	Post Test (f)
Not Skilled	Lacks foundational math skills	0 – 5	1	0
Limited Skilled	Shows partial understanding of basic mathematical concepts	6 – 11	6	2
Moderately Skilled	Meets minimum competency; demonstrates basic mathematical literacy	12	2	3
Skilled	Demonstrates solid understanding and application of mathematical skills	13 – 15	7	10
Highly Skilled	Demonstrates advanced understanding and mastery of mathematical literacy	16 – 20	26	27
Total			42	42

Table 10 presents the distribution of kindergarten pupils according to their skill levels in overall mathematical literacy. Prior to the intervention, the largest group of pupils was classified as "Highly Skilled" (26 pupils, 61.9%), followed by "Skilled" (7 pupils, 16.7%), "Limited Skilled" (6 pupils, 14.3%), and "Moderately Skilled" (2 pupils, 4.8%). One pupil (2.4%) was classified as "Not Skilled," lacking foundational math skills.



Following the intervention, the number of pupils in the "Highly Skilled" category increased from 26 to 27 (64.3%), representing a slight but meaningful increase. More notably, the "Not Skilled" category was reduced to 0, and the "Limited Skilled" category decreased from 6 to 2 (4.8%). The "Skilled" category increased from 7 to 10 (23.8%), while the "Moderately Skilled" category increased slightly from 2 to 3 (7.1%). These changes indicate an overall upward shift in the distribution of skill levels, with the greatest improvements observed in the lower skill categories.

The reduction of the "Limited Skilled" category from 6 to 2 represents a 66.7% decrease in the number of pupils demonstrating only partial understanding of mathematical concepts. Similarly, the complete elimination of the "Not Skilled" category indicates that the intervention successfully addressed the needs of the lowest-performing pupil. These findings are particularly significant in light of the research by Nguyen et al. (2016), which demonstrated that early mathematics deficits tend to compound over time if left unaddressed. By intervening at the kindergarten level, this study demonstrated that it is possible to close foundational learning gaps before they become more pronounced.

The increase in the "Skilled" and "Highly Skilled" categories suggests that the intervention not only helped struggling learners but also supported proficient learners in advancing to higher skill levels. This finding aligns with the Learning Trajectories Approach proposed by Clements and Sarama (2021), which



emphasizes the importance of providing instructional materials that scaffold learning for children at different developmental levels. The intervention's effectiveness across different skill levels supports the use of diagnostic-intervention cycles that assess current abilities and provide targeted instruction to bridge gaps between current and potential performance.

Table 11

Significant difference between the levels of overall mathematical literacy in terms of age

		Statistic	p		Effect Size	Interpretation
PRETEST OVERALL	Mann-Whitney U	171	0.210	Rank biserial correlation	0.2227	No significance
POSTTEST OVERALL	Mann-Whitney U	212	0.839	Rank biserial correlation	0.0386	No significance

The analysis investigated whether significant differences exist in the levels of overall mathematical literacy based on the age of the participants. Table 11 presents the results of the Mann-Whitney U test for both the pretest and posttest phases. Regarding the Pretest Overall scores, the results indicated no statistically significant difference between age groups ($U = 171$, $p = .210$). The calculated effect size, using Rank-biserial correlation, was 0.2227, which suggests a small to



nearing-moderate relationship, yet it does not reach the threshold for statistical significance. This implies that at the beginning of the study, participants demonstrated comparable levels of mathematical literacy regardless of their age.

Similarly, the Posttest Overall results showed no significant difference in mathematical literacy levels in terms of age ($U = 212$, $p = .839$). The effect size for the posttest was notably low ($r_{rb} = 0.0386$), indicating a negligible relationship between age and the outcomes following the intervention. These findings suggest that the intervention was equally effective across the different age groups represented in the sample, as the post-intervention literacy levels remained statistically uniform.

In summary, age was not a determining factor in the mathematical literacy performance of the participants, either before or after the intervention. The high p -values ($p > .05$) in both instances lead to the conclusion that any observed differences in scores across age groups were likely due to chance rather than a systemic influence of age. This finding supports the null hypothesis that there is no significant difference in mathematical literacy levels in terms of age.

The lack of significant age differences in mathematical literacy among kindergarteners in this study may be attributed to several factors. First, the age range within the sample was limited (5 to 6 years old), which may not represent sufficient developmental variation to produce meaningful differences. Second, the kindergarten curriculum in the Philippines is designed to provide foundational



mathematical experiences that are appropriate for both age groups, potentially minimizing any developmental advantages that older children might have. Third, as noted by Geary (2018), early mathematical development is influenced by a complex interaction of cognitive, environmental, and instructional factors, with age being only one of many contributing variables.

Table 12

Significant difference between the levels of overall mathematical literacy before and after the intervention

		Statistic	p		Effect Size	Interpretation
PRETEST OVERALL	Mann-Whitney U	149	0.068	Rank biserial correlation	0.32273	No significance
POSTTEST OVERALL	Mann-Whitney U	219	0.990	Rank biserial correlation	-0.00455	No significance

The analysis evaluated whether a significant difference exists in the participants' levels of overall mathematical literacy when grouped by sex. Table 12 details the results of the Mann-Whitney U test for both pre-intervention and post-intervention phases. Regarding the Pretest Overall mathematical literacy, the results indicated no statistically significant difference between male and female participants ($U = 149$, $p = .068$). Although the p-value is approaching the alpha



threshold of .05, it remains non-significant. The Rank-biserial correlation of .323 suggests a moderate effect size, indicating that while there was some variance in baseline scores between sexes, it was not statistically sufficient to conclude a systemic difference.

For the Posttest Overall results, the difference between sexes became even less pronounced ($U = 219$, $p = .990$). The p -value, being nearly 1.00, indicates that the distribution of scores between males and females was almost identical following the intervention. Furthermore, the effect size reached a negligible value ($r_{rb} = -0.005$), confirming that sex had no impact on the mathematical literacy outcomes after the program was conducted.

The data demonstrates that mathematical literacy levels were not significantly influenced by sex at either stage of the study. This finding supports the null hypothesis that there is no significant difference in mathematical literacy levels in terms of sex.

The absence of significant gender differences in mathematical literacy among kindergarteners in this study aligns with contemporary research on gender and mathematics. Hyde et al. (2019) argued that the "gender gap" in mathematics has been overstated and is largely absent in early childhood, with differences only emerging later due to sociocultural factors rather than inherent cognitive differences. Similarly, the findings of this study suggest that male and female



kindergarteners in this sample demonstrated comparable mathematical literacy levels both before and after the intervention.

The moderate effect size observed in the pretest ($r_{rb} = 0.323$) warrants some consideration, as it suggests that there may have been slight differences in baseline scores between sexes that were not large enough to reach statistical significance given the sample size. However, the posttest effect size of nearly zero indicates that the intervention effectively eliminated even these minor differences, resulting in nearly identical performance between male and female participants. This finding supports the effectiveness of the intervention in promoting equitable learning outcomes regardless of sex.

Table 13

			Statistic	p		Effect Size	Interpretation
PRETEST OVERALL	POSTTEST OVERALL	Wilcoxon W	143 ^a	0.008	Rank biserial correlation	-0.519	Significance

The analysis revealed a statistically significant difference between the pretest and posttest overall scores ($W = 143$, $p = .008$). Because the p-value is less than the conventional alpha level of .05, the null hypothesis which posits no change in literacy levels is rejected. This finding indicates that the change in mathematical literacy from the pretest phase to the posttest phase is unlikely to have occurred by chance.



The Rank-biserial correlation was calculated at -0.519 . In the context of the Wilcoxon test, an effect size of this magnitude (approximately 0.52) is considered large. The negative value indicates the direction of the ranks, confirming that posttest scores were consistently higher than pretest scores. This large effect size suggests a substantial and practical improvement in the participants' mathematical literacy over time.

Therefore, there was a significant and meaningful improvement in mathematical performance. The transition of the mean score from 14.79 (Skilled) to 16.02 (Highly Skilled), coupled with a significant p -value and a large effect size, provides strong evidence for the improvement in foundational mathematical skills.

The significant improvement observed in overall mathematical literacy is consistent with the findings of Bailey et al. (2020), who demonstrated that early childhood mathematics gains can show meaningful changes in mathematical outcomes. The large effect size (0.52) indicates that the change was not only statistically significant but also practically meaningful, representing a substantial improvement in performance relative to the variability in scores.

The improvement can be attributed to several factors. First, the use of a diagnostic cycle, as outlined in the conceptual framework, allowed for a better understanding of the specific needs of learners. Second, the instructional activities (or the regular classroom instruction) were designed to be developmentally appropriate, aligning with the Learning Trajectories Approach proposed by



Clements and Sarama (2021). Third, the assessment covered multiple domains of mathematical literacy (color and shape recognition, number recognition and counting, and readiness for basic operations), providing a comprehensive picture of early mathematical development.

The finding that there was significant improvement despite the already high baseline performance of participants is particularly noteworthy. The pretest median of 16.00 indicated that more than half of the participants were already performing at the "Highly Skilled" level before the pretest. Nevertheless, the significant improvement in mean scores and the upward shift in the distribution of skill levels demonstrate that even learners with strong foundational skills can show further gains over time. This finding supports the notion that early mathematical development is not a fixed trajectory but can be enhanced through appropriate instruction and scaffolding.



Chapter 4

SUMMARY OF FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of the study, the findings that were revealed in this study, the conclusions that were drawn from the findings and the recommendations formulated by the researcher.

Summary of Findings

This study aimed to get a clearer picture of how kindergarten pupils at New Cabalan Elementary School were doing in terms of early math skills and to see how those skills changed after going through their regular classroom instruction. Here's what we found:

We had 42 kindergarten pupils in total. In terms of age, 22 were five years old (about 52%) and 20 were six years old (about 48%). As for sex, 22 were girls (52%) and 20 were boys (48%). So, a nice balance across both age and sex.

We looked at the main three areas which are the color/shape recognition (including sequencing), number recognition and counting, and readiness for basic operations (addition and subtraction) showed that children began the intervention with strong foundational skills, with most already in the "Skilled" or "Highly Skilled" ranges. Following instruction, performance improved across all areas, the number of children reaching "Highly Skilled" in color/shape recognition doubled from six to twelve, while struggling learners dropped from four to one. Number recognition



and counting saw the most notable progress, with mastery increasing from sixteen to twenty-two children, and the two lowest-performing students moved entirely out of the beginner category. Readiness for basic operations was high from the start (thirty-six of forty-two children already "Highly Skilled"), and by the posttest, the remaining students caught up, leaving no one behind. Importantly, comparisons by age (five- vs. six-year-olds) and gender revealed no significant differences in performance or growth before or after the intervention, indicating that both age groups and both boys and girls benefited equally from classroom instruction.

This was the heart of the study. When we compared the kids' overall math literacy before and after intervention, the improvement was clear and meaningful. The average score went up, the number of kids achieving mastery increased, and the kids who were struggling the most showed the biggest leaps. The difference was statistically significant, and the effect was large meaning this wasn't just a small change, but a real, substantial improvement.

Conclusions

The study concluded that kindergarten pupils at New Cabalan Elementary School demonstrated a high level of mathematical literacy even before the intervention, particularly in readiness for basic operations, number recognition, and color and shape recognition. When compared according to age, no significant



differences were found in the pupils' overall mathematical literacy, indicating that five- and six-year-olds performed similarly. Likewise, when compared according to sex, no significant differences were observed between male and female pupils, showing that both groups had comparable mathematical abilities. However, when the pretest and posttest overall mathematical literacy levels were compared, a significant difference was found, indicating meaningful improvement following the regular classroom instruction. These findings confirmed that while demographic factors such as age and sex did not influence mathematical literacy, the teaching and learning process played a key role in enhancing kindergarteners' foundational math skills.

Recommendations

Based on the findings of this study, the following recommendations are offered:

1. For Teachers, continue using simple, developmentally appropriate assessments like the one in this study to regularly check where children are and track their growth over time. Knowing where each child stands makes it easier to provide the right support.

Build on the strengths already present in the classroom. Since many children came in with solid foundational skills, consider offering slightly more challenging activities for those ready to go further, while still supporting those who need more practice.



Keep using hands-on, play-based activities that naturally build number sense, shape recognition, and counting skills. The growth seen in this study shows that what teachers are already doing is working.

2. For School Administrators, support kindergarten teachers by providing resources such as manipulatives, counting materials, and shape-related learning tools that make math learning engaging and concrete for young children.

Encourage and protect time for teachers to do simple pre- and post-assessments, as these give valuable insights into student progress and help guide instructional decisions.

Consider establishing a "Math Tutoring Thursday" program, a dedicated day each week where kindergarteners who need a little extra support can receive focused, one-on-one or small-group help in a friendly and encouraging setting. This gives struggling learners the chance to catch up without feeling left behind.

3. For Parents and Guardians, share the findings of this study with parents to reassure them that their children are growing in math skills through regular classroom learning. Encourage them to continue supporting math



learning at home through simple activities like counting objects, identifying shapes around the house, and playing number games.

Parents may also explore the use of educational tools such as the newly developed "MATHVENTURE" math application tailored specifically for Filipino kindergarteners, which can provide additional practice in a fun and engaging way.

4. For the Department of Education, the strong baseline skills and growth observed in this study suggest that early childhood math instruction in this context is on the right track. Consider using similar simple assessment tools more widely to better understand the state of early numeracy across different schools.

Continue supporting kindergarten teachers with training and resources focused on foundational math skills, as these early years are critical for later academic success.

Provide opportunities for the pilot testing and integration of locally developed educational technology, such as the "MATHVENTURE" math application created for Filipino kindergarteners, into classroom instruction or remedial programs.



5. For Future Researchers, consider expanding this study to include more schools and a larger number of participants to see if similar patterns of growth hold true in other settings.

Explore other domains of early mathematical literacy, such as patterning, spatial reasoning, or measurement, to get a fuller picture of young children's math development.

Conduct follow-up studies to see how kindergarten math skills relate to later performance in Grade 1 and beyond.

Investigate the effectiveness of programs like Math Tutoring Thursday and locally developed math applications in supporting early numeracy development, particularly for learners who struggle with foundational skills.



BIBLIOGRAPHY

A. BOOKS

- Clements, D. H., & Sarama, J. (2021). *Learning and teaching early math: The learning trajectories approach* (3rd ed.). Routledge.
<https://www.routledge.com/Learning-and-Teaching-Early-Math/Clements-Sarama/p/book/9780367521974>
- National Association for the Education of Young Children (NAEYC). (2020). *Developmentally appropriate practice in early childhood programs serving children from birth through age 8* (4th ed.).
<https://www.naeyc.org/resources/pubs/books/developmentally-appropriate-practice-4th-edition>
- National Research Council. (2018). *How people learn II: Learners, contexts, and cultures*. The National Academies Press.
<https://nap.nationalacademies.org/catalog/24783>
- OECD. (2019). *PISA 2018 Results (Volume I): What Students Know and Can Do*. OECD Publishing. <https://www.oecd.org/pisa/publications/pisa-2018-results-volume-i-5f07c754-en.htm>
- Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Harvard University Press.
<https://www.hup.harvard.edu/catalog.php?isbn=9780674576292>

B. JOURNALS

- Bailey, D. H., Duncan, G. J., Watts, T., Clements, D. H., & Sarama, J. (2020). The long-term impacts of early childhood mathematics intervention. *Journal of Research on Educational Effectiveness*, 13(4), 705–730. <https://doi.org/10.1080/19345747.2020.1752444>
- Lima, F. A. de, & Sousa, J. B. (2025). *Institutional change and endogenous development: Theoretical contributions*. *Economies*, 13(6), 165.
<https://doi.org/10.3390/economies13060165>



- Bernardo, A. B. I. (2021). *Exploring the challenges and opportunities in Mother Tongue-Based Multilingual Education in the Philippines*. Philippine Journal of Psychology. <https://pap.ph/resources/education-research/>
- Chang, A., Sandhofer, C. M., & Brown, C. S. (2020). Gender biases in early number exposure to boys and girls. *Journal of Experimental Child Psychology*, 192, 104774. <https://doi.org/10.1016/j.jecp.2019.104774>
- Geary, D. C. (2018). The nature of mathematical cognitive development. *Journal of Numerical Cognition*, 4(1), 1–17. <https://doi.org/10.5964/jnc.v4i1.161>
- Ginsburg, H. P., Lee, J. S., & Boyd, J. S. (2019). Mathematics education for young children: What it is and how to promote it. *Social Policy Report*, 32(1), 1-24. <https://srcd.onlinelibrary.wiley.com/doi/10.1002/j.2379-3988.2019.tb00001.x>
- Hyde, J. S., Bigler, R. S., Joel, D., Tate, C. C., & van Anders, S. M. (2019). The future of sex and gender in psychology: Five challenges to the gender binary. *American Psychologist*, 74(2), 171–193. <https://doi.org/10.1037/amp0000307>
- Leyva, D. (2019). How do low-income Chilean parents support their preschoolers' mathematical development? *Journal of Early Childhood Literacy*, 19(4), 493–519. <https://doi.org/10.1177/1468798417745118>
- Lim, S. M. (2022). *Pedagogical Content Knowledge in Early Mathematics: A Study of Philippine Kindergarten Teachers*. International Journal of Early Childhood Education.
- Nguyen, T., Watts, T. W., Duncan, G. J., Clements, D. H., Sarama, J., Wolfe, C., & Spitler, M. E. (2016). Which preschool mathematics competencies are most predictive of fifth grade achievement? *Early Childhood Research Quarterly*, 36, 550–560. <https://doi.org/10.1016/j.ecresq.2016.02.006>
- Peres, F.F. (2025). Effect sizes for nonparametric tests. *Biochemia Medica vol. 36*. <https://doi.org/10.11613/BM.2026.010101>
- Purpura, D. J., & Lonigan, C. J. (2016). Early numeracy assessment: The development of the Preschool Numeracy Scales. *Early Education and Development*, 27(2), 205–228. <https://doi.org/10.1080/10409289.2016.1102017>



Purpura, D. J., Logan, J. A. R., Hassinger-Das, B., & Napoli, A. R. (2017). Why do early mathematics skills predict later reading? The role of mathematical language. *Developmental Psychology*, 53(9), 1633–1642.
<https://doi.org/10.1037/dev0000343>

Watts, T. W., Duncan, G. J., Clements, D. H., & Sarama, J. (2018). What is the long-run impact of learning mathematics during preschool? *Child Development*, 89(2), 539–555. <https://doi.org/10.1111/cdev.12731>

Zippert, E. L., & Rittle-Johnson, B. (2020). The home math environment: More than numeracy. *Early Childhood Research Quarterly*, 50, 4–15.
<https://doi.org/10.1016/j.ecresq.2019.12.006>

C. ELECTRONIC SOURCES

The Field of Facilitation. (2017). *Facilitator practice: the intervention cycle*.
<https://thefieldoffacilitation.wordpress.com/2017/08/15/facilitator-practice-the-intervention-cycle/>



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



APPENDICES



Appendix A

Letter of permission to conduct the study



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



December 2, 2025


Darwin P. Paguio, PhD.
Vice President of Academic Affairs
Dean, College of Education, Arts and Sciences
Gordon College

Dear Sir,

We hope this message finds you well. We are Bachelor of Early Childhood Education students writing to seek your approval to conduct a research study titled **"Counting the Uncounted: Exploring the Mathematical Literacy Levels among Kindergarteners in New Cabalan Elementary School."**

The purpose of this research is to assess the mathematical literacy levels of kindergarten students at New Cabalan Elementary school. We aim to understand their current skills and identify any areas where additional support may be needed to enhance their learning outcomes. This study is essential not only for understanding the educational landscape of our young learners but also for informing potential interventions that could benefit their development.

We assure you that the research will be conducted ethically and with the utmost respect for the students and the school community. Participation will be voluntary, and all data collected will remain confidential.

We would greatly appreciate your support and approval to proceed with this important study. If you require any further information or would like to discuss this proposal in more detail, please feel free to contact us at 202310906@gordoncollege.edu.ph or 09154995033 at your convenience.

Thank you for considering this request. I look forward to your positive response.



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



Sincerely,

[Redacted Signature]

Guienn D. Garganta
3rd Year BECEd Student

[Redacted Signature]

Alyssa Rica Librero
3rd Year BECEd Student

[Redacted Signature]

Donna May J. Mesina
3rd Year BECEd Student

[Redacted Signature]

Georgia Victora Villafania
3rd Year BECEd Student

Noted by:

[Redacted Signature]

Rachelle Ann D. Ignacio, LPT
Research Adviser

[Redacted Signature]

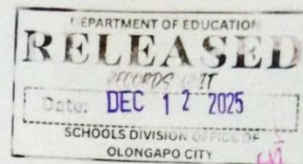
Anna Theresa M. Amarille, MASE
Assistant Dean - CEAS



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
Department of Education
Region III
SCHOOLS DIVISION OFFICE OF OLONGAPO CITY



OFFICE OF THE SCHOOLS DIVISION SUPERINTENDENT

04 DECEMBER 2025

GUIENN D. GARGANTA Et. al
Researcher, BS Early Childhood Education
Gordon College, Olongapo City

Dear Ms. Garganta,

Magandang Buhay! Mabuting Tao!

This is to inform you that your request to conduct your study titled "Counting the Uncounted: Exploring the Mathematical Literacy Levels among Kindergarteners", among teachers and kindergarten learners at New Cabalan Elementary School, is **approved**.

In line with this, it is firmly requested that the following provisions should be followed, to wit:

- schedule in the administration of tests is on the discretion of school heads/principals;
- the activity should not entail additional cost to the schools;
- no collection policy should be strictly observed;
- participation of respondents is on voluntary basis;
- secure parents' consent for every learner who will undergo the testing;
- the conduct of survey should not disrupt classes and/or school operation;
- information gathered should be used solely for research purposes;
- no government resources should be used in this endeavor; and
- submit to this office a copy of the results and findings.

Very truly yours,

IMELDA P. MACASPAC PhD
Schools Division Superintendent

msb/120425/SDOOC-CID-DIA



Address: Gordon Avenue, New Asinan, Olongapo City
Contact No.: 222-2568/ 603-0570
Email Address: olongapo.city@deped.gov.ph
Official Website: <https://deped-olongapo.com>
"SDO Olongapo City: Towards a Culture of Excellence and Character"



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos. : (047) 224-2089/(047) 602-7175



Sincerely,

[Redacted Signature]

Guinn D. Garganta
3rd Year BECED Student

[Redacted Signature]

Alyssa Rica Librero
3rd Year BECED Student

[Redacted Signature]

Donna May J. Mesina
3rd Year BECED Student

[Redacted Signature]

Georgia Victoria Villafania
3rd Year BECED Student

Noted by:

[Redacted Signature]

Rachelle Ann D. Ignacio, LPT
Research Adviser

[Redacted Signature]

Anna Theresa M. Amarille, MASE
Assistant Dean - CEAS

[Redacted Signature]

Dr. Darwin P. Paguio
Vice President of Academic Affairs
Dean CEAS



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



January 6, 2026

Marites Calara, EdD
School Principal
New Cabalan Elementary School
Olongapo City

Dear Madame;

Good day! We are Third Year Bachelor of Early Childhood Education students of Gordon College Olongapo City humbly seek the permission from your good office to allow us to administer a pre-test and post-test to the morning and afternoon kindergarten classes of Mrs. Leonita Flojo for our requirements in the course EDUECE19a – Research in Early Childhood Education for the Second Semester, AY 2025 – 2026 titled “Counting the Uncounted: Exploring the Mathematical Literacy Levels among Kindergarteners in New Cabalan Elementary School.”

The purpose of this research is to assess the mathematical literacy levels of kindergarten students at New Cabalan Elementary school. We aim to understand their current skills and identify any areas where additional support may be needed to enhance their learning outcomes. This study is essential not only for understanding the educational landscape of our young learners but also for informing potential interventions that could benefit their development.

We assure you that the research will be conducted ethically and with the utmost respect for the students and the school community. Participation will be voluntary, and all data collected will remain confidential.

For your reference, we have attached the approved letter from the Division Office, along with the validated questionnaires and the Table of Specifications.

Received:


HM/DIC-ASP

01-06-2025

(0928408196)



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



We would greatly appreciate your support and approval to proceed with this important study. If you require any further information or would like to discuss this proposal in more detail, please feel free to contact us at 202310906@gordoncollege.edu.ph or 09154995033 at your convenience.

Thank you for considering this request. We look forward to your positive response.

Sincerely,

[Redacted Signature]

Guinn D. Garganta
3rd Year BECEd Student

[Redacted Signature]

Alyssa Rica Librero
3rd Year BECEd Student

[Redacted Signature]

Donna May J. Mesina
3rd Year BECEd Student

[Redacted Signature]

Georgia Victoria Villafania
3rd Year BECEd Student

Noted by:

[Redacted Signature]

Rachelle Ann D. Ignacio, LPT
Research Adviser

[Redacted Signature]

Anna Theresa M. Amarille, MASE
Assistant Dean - CEAS

[Redacted Signature]

Dr. Eder E. Magsayo
Dean - CEAS



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



January 6, 2026

Mrs. Leonita Flojo
Teacher II
New Cabalan Elementary School
Olongapo City

Dear Madam,

Good day! We are future early childhood educators from Gordon College, currently conducting our research titled, **"Counting the Uncounted: Exploring the Mathematical Literacy Levels among Kindergarteners in New Cabalan Elementary School,"** aims to better understand early math skills. We are excited by the possibility of partnering with New Cabalan Elementary School to conduct this meaningful work.

We respectfully seek your permission to conduct our research with the kindergarten pupils under your supervision. The study aims to assess their mathematical literacy levels, focusing on their counting skills and identifying areas where further support may enhance their learning development.

Before the actual conduct of the research, we will be distributing consent forms to the parents or guardians of the pupils to ensure that their participation is voluntary and fully informed.

Rest assured that the research will be carried out with professionalism, respect, and care for the children and the school community. All information gathered will remain confidential and will be used solely for academic purposes.

We sincerely appreciate your time and kind cooperation in supporting the completion of our study. Attached here are the validated questionnaires and the Table of Specifications. If you require any further information or would like to discuss this proposal in more detail, please feel free to contact us at 202310906@gordoncollege.edu.ph or 09154995033 at your convenience.



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



Respectfully yours,


Guienn D. Garganta

3rd Year BECED Student


Alyssa Riza Librero

3rd Year BECED Student


Donna May J. Mesina

3rd Year BECED Student


Georgia Victora Villafania

3rd Year BECED Student

Noted by:


Rachelle Ann B. Ignacio, LPT

Research Adviser


Anna Theresa M. Amarille, MASE

Assistant Dean - CEAS


Dr. Eder E. Magsayo

Dean - CEAS



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



January 6, 2026

PARENTAL CONSENT FORM

TO THE PARENTS/GUARDIANS OF KINDERGARTEN PUPILS:

We are students from the Bachelor of Early Childhood Education (BECEd) program at Gordon College. As part of our academic requirements, we are conducting a research study entitled: **"Counting the Uncounted: Exploring the Mathematical Literacy Levels among Kindergarteners in New Cabalan Elementary School."**

The purpose of this study is to better understand the early mathematical skills, particularly counting, of kindergarten pupils. This will help identify their current levels and areas where supportive learning strategies can be developed.

We are writing to respectfully request your permission for your child to participate in this study.

If you consent, your child will participate in an administered age appropriate test with a researcher to assess basic counting skills. This activity will be conducted during school hours in a familiar and comfortable setting within the school, under the supervision of their teacher. The activity will take approximately 15-20 minutes.

Your child's privacy is our top priority. All information gathered will remain strictly confidential. Your child's name or personal identifiers will not appear in any part of the research report, presentations, or publications. Data will be reported only as group findings or anonymous summaries. All records will be kept secure and destroyed after the academic evaluation of the study.

Participation in this study is completely voluntary. Your child has the right to decline or stop the activity at any time without any penalty or negative consequences to their academic standing or relationship with the school or teachers.



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
COLLEGE OF EDUCATION, ARTS, AND SCIENCES
Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo City
Tel Nos.: (047) 224-2089/(047) 602-7175



If you have any questions or concerns about the study, please feel free to contact us at:
202310906@gordoncollege.edu.ph or 0915-499-5033.

Respectfully yours,

[Redacted]
Guienn P. Garganta

3rd Year BECED Student

[Redacted]
Alyssa Riza Librero

3rd Year BECED Student

[Redacted]
Donna May J. Mesina

3rd Year BECED Student

[Redacted]
Georgia Victoria Villafania
3rd Year BECED Student

Noted by:

[Redacted]
Rachelle Ann D. Ignacio, LPT
Research Adviser



Appendix B Questionnaire

The Instrument



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
Olongapo City Sports Complex, Donor St., East Tapinac, Olongapo City
COLLEGE OF EDUCATION, ARTS AND SCIENCES



KINDERGARTEN PRE-TEST IN NUMERACY

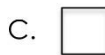
PANGALAN: _____

PANUTO: Bilugan ang letra nang tamang sagot.

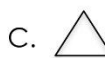
1. Alin sa mga sumusunod ang kulay asul?



2. Alin sa mga sumusunod ang hugis parisukat?



3. Alin ang pinakamalaki?



4. Alin ang pinakamaliit?



5. Piliin kung alin sa mga sumusunod ang nagsisimula sa pinakamaliit hanggang pinakamalaki.



6. Piliin kung alin sa mga sumusunod ang nagsisimula sa pinakamalaki hanggang pinakamaliit.





7. Ilang beses naulit ang letrang **"e"** sa salitang **elepante**?

A. 1

B. 3

C. 5

8. Ilang lahat ang okra ng nakikita mo?



A. 5

B. 4

C. 3

9. Ilang lahat ang krayola ng nakikita mo?



A. 5

B. 6

C. 7

10. Ilang ang nagpapakita ng masayang mukha?



A. 4

B. 5

C. 7

11. Anong bilang ang susunod sa **8**?

A. 7

B. 9

C. 10

12. Anong bilang ang mauuna sa **4**?

A. 3

B. 5

C. 6



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



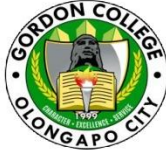
Republic of the Philippines
City of Olongapo
GORDON COLLEGE
Olongapo City Sports Complex, Donor St., East Tapinac, Olongapo City
COLLEGE OF EDUCATION, ARTS AND SCIENCES



13. Anong bilang ang nasa pagitan ng 10 at 12?
- A. 9 B. 11 C. 13
14. Alin sa mga sumusunod na kahon ang may bilang mula sa pinakamaliit hanggang pinakamalaki?
- A. B. C.
15. Alin sa mga sumusunod na kahon ang may bilang mula sa pinakamalaki hanggang pinakamaliit?
- A. B. C.
16. Alin sa mga sumusunod ang 10:00 o'clock?
- A.  B.  C. 
17. Alin sa mga sumusunod ang 12:00 o'clock?
- A.  B.  C. 
18. Alin sa mga sumusunod ang 5:00 o'clock?
- A.  B.  C. 



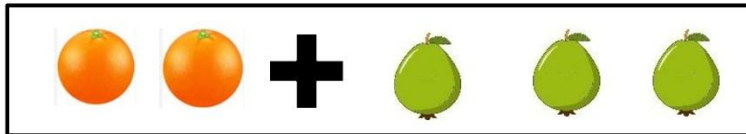
Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
Olongapo City Sports Complex, Donor St., East Tapinac, Olongapo City
COLLEGE OF EDUCATION, ARTS AND SCIENCES

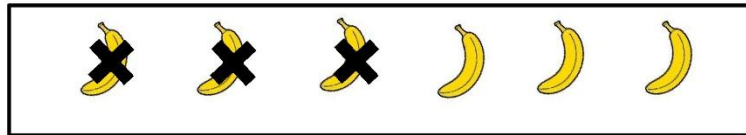


19. Si Alyssa ay mayroong 2 kahel. Binigyan siya ng nanay niya ng 3 bayabas. Ilang prutas meron lahat si Alyssa?



- A. 4 B. 5 C. 6

20. Si Georgia ay may 6 na saging. Kinain niya yung 3 na saging. Ilang nalang ang saging ni Georgia na natira?



- A. 3 B. 5 C. 6



KINDERGARTEN POST-TEST IN NUMERACY

PANGALAN: _____

PANUTO: Bilugan ang letra nang tamang sagot.

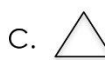
1. Alin sa mga sumusunod ang kulay asul?



2. Alin sa mga sumusunod ang hugis parisukat?



3. Alin ang pinakamalaki?



4. Alin ang pinakamaliit?



5. Piliin kung alin sa mga sumusunod ang nagsisimula sa pinakamaliit hanggang pinakamalaki.



6. Piliin kung alin sa mga sumusunod ang nagsisimula sa pinakamalaki hanggang pinakamaliit.





7. Ilang beses naulit ang letrang **"e"** sa salitang **elepante**?

A. 1

B. 3

C. 5

8. Ilan lahat ang okra ng nakikita mo?



A. 5

B. 4

C. 3

9. Ilan lahat ang krayola ng nakikita mo?



A. 5

B. 6

C. 7

10. Ilan ang nagpapakita ng masayang mukha?



A. 4

B. 5

C. 7

11. Anong bilang ang susunod sa **8**?

A. 7

B. 9

C. 10

12. Anong bilang ang mauuna sa **4**?

A. 3

B. 5

C. 6



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
Olongapo City Sports Complex, Donor St., East Tapinac, Olongapo City
COLLEGE OF EDUCATION, ARTS AND SCIENCES



13. Anong bilang ang nasa pagitan ng 10 at 12?

A. 9

B. 11

C. 13

14. Alin sa mga sumusunod na kahon ang may bilang mula sa pinakamaliit hanggang pinakamalaki?

A. 2, 3, 4

B. 3, 2, 4

C. 4, 3, 2

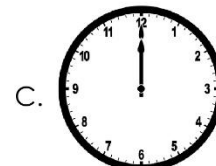
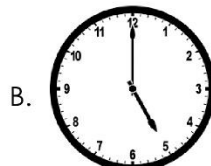
15. Alin sa mga sumusunod na kahon ang may bilang mula sa pinakamalaki hanggang pinakamaliit?

A. 6, 5, 7

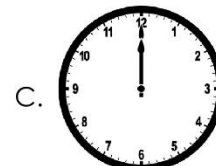
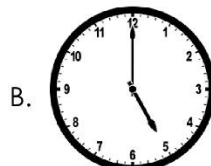
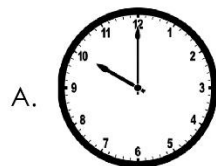
B. 5, 6, 7

C. 7, 6, 5

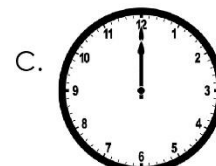
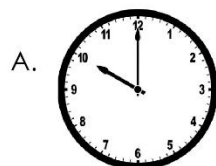
16. Alin sa mga sumusunod ang 10:00 o'clock?



17. Alin sa mga sumusunod ang 12:00 o'clock?

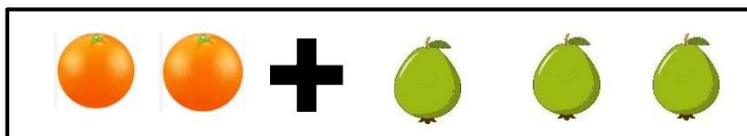


18. Alin sa mga sumusunod ang 5:00 o'clock?



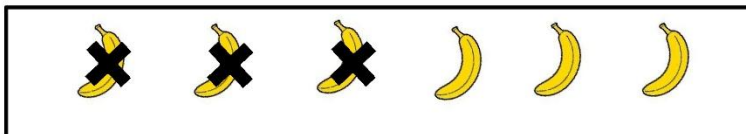


19. Si Alyssa ay mayroong 2 kahel. Binigyan siya ng nanay niya ng 3 bayabas. Ilang prutas meron lahat si Alyssa?



- A. 4 B. 5 C. 6

20. Si Georgia ay may 6 na saging. Kinain niya yung 3 na saging. Ilang nalang ang saging ni Georgia na natira?



- A. 3 B. 5 C. 6



Numeracy
 Subject

Kindergarten
 Grade

Table of Specifications
 1st to 4th
 Grading Period

2025-2026
 School Year

Competencies	Time Spent/ Freq.	No. of Items	Percentage	Level of Cognition				
				Remembering POI	Understanding POI	Applying POI	Analyzing POI	Evaluating POI
Sort and classify objects according to one attribute/property (shape, color, size).	4	4	20%	1, 2, 3, 4	2	3, 4		
Arrange objects one after another in a series/sequence according to given attribute (size, length) and describe their relationship (big/bigger/biggest or long/longer/longest).	2	2	10%		5, 6			5, 6
Count objects with one-to-one correspondence up to quantities of 10.	4	4	20%	7, 8, 9, 10		8, 9		
Tell time by the hour.	3	3	15%			11, 12, 13		
Identify the number that comes before, after, or in between.	3	3	15%		14, 15, 16			
Arrange three numbers from least to greatest/greatest to least.	2	2	10%			17, 18		
Recognize the words "put together," "add to," and "in all" that indicate the act of adding whole numbers.	1	1	5%				19	
Recognize the words "take away," "less," and "are left" that indicate the act of subtracting whole numbers.	1	1	5%				20	
TOTAL:	20 minutes	20	100%					

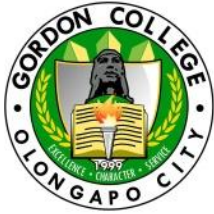


Appendix C Normality test

Descriptives

	Mean	Median	Shapiro-Wilk	
			W	p
CSS 1	4.60	5.00	0.881	<.001
NRC 1	8.40	9.00	0.842	<.001
RBO 1	1.79	2.00	0.424	<.001
PRETEST OVERALL	14.79	16.00	0.836	<.001
CSS 2	4.93	5.00	0.846	<.001
NRC 2	9.24	10.00	0.925	0.009
RBO 2	1.86	2.00	0.417	<.001
POSTTEST OVERALL	16.02	16.00	0.947	0.053

Descriptive statistics and tests of normality for learners' mathematical literacy are presented in Table. In the pretest, learners scored a mean of 4.60 and median of 5.00 in Color and Shape Recognition and Sequencing (CSS 1), 8.40 and 9.00 in Number Recognition and Counting (NRC 1), and 1.79 and 2.00 in Readiness for Basic Operations (RBO 1). The overall pretest mean and median were 14.79 and 16.00, respectively. Shapiro-Wilk tests indicated that all pretest scores were significantly non-normally distributed (CSS 1: $W = 0.881$, $p < .001$; NRC 1: $W = 0.842$, $p < .001$; RBO 1: $W = 0.424$, $p < .001$; Pretest Overall: $W = 0.836$, $p < .001$), supporting the use of nonparametric statistical analyses.



For the posttest, learners obtained means and medians of 4.93 and 5.00 for CSS 2, 9.24 and 10.00 for NRC 2, and 1.86 and 2.00 for RBO 2. The overall posttest mean and median were 16.02 and 16.00, respectively. Shapiro-Wilk results showed that most posttest scores remained non-normally distributed (CSS 2: $W = 0.846$, $p < .001$; NRC 2: $W = 0.925$, $p = .009$; RBO 2: $W = 0.417$, $p < .001$), although the overall posttest approached normality ($W = 0.947$, $p = .053$). These findings indicate that learners generally demonstrated improvement in mathematical literacy from pretest to posttest, with median scores often slightly higher than means, suggesting that while most learners performed at or above the skill level indicated, a few lower scores slightly reduced the averages. The violation of normality in most domains justifies the use of nonparametric tests in subsequent inferential analyses.



APPENDIX D

INTERPRETATION OF SCORES:

Table 1

Color, Shape Recognition, and Sequencing

Raw Score Range	% Range	Skill level	Interpretation
0 – 1	60 – 66.67	Not Yet Skilled	Cannot recognize colors/shapes; no sequencing skill
2 – 3	70 – 74	Low or Limited skill	Low – Limited recognition; sequencing needs assistance
4	75 – 80	Moderately Skilled	Basic recognition and sequencing achieved
5	85 – 90	Skilled	Good recognition and sequencing skills
6	95 – 100	Highly Skilled	Mastery of recognition and sequencing

Table 2

Number Recognition and Counting

Raw Score Range	% Range	Skill Level	Interpretation
0 – 2	60 – 66.67	Not Skilled	Cannot recognize numbers or count reliably
3 – 6	70 – 74	Limited Skilled	Recognizes some numbers and counts with assistance



7	75 – 80	Moderately Skilled	Recognizes numbers and counts with basic accuracy
8 – 9	83 – 88	Skilled	Accurately recognizes numbers and counts with minimal errors
10 – 12	90 – 100	Highly Skilled	Demonstrates mastery in number recognition and counting

Table 3

Readiness for Basic Operations

Raw Score Range	% Range	Skill Level	Interpretation
0	60	Not Yet Skilled	No understanding of operations
1	75 – 80	Skilled	Emerging readiness for operations
2	95 – 100	Highly Skilled	Ready for basic operations independently



APPENDIX E

CERTIFICATE OF STATISTICIAN

CERTIFICATE OF DATA ANALYSIS

This is to certify that the undersigned has conducted the statistical treatment, and analysis of data for the research study entitled:

“Counting The Uncounted: Exploring The Mathematical Literacy Levels Among Kindergarteners In New Cabalan Elementary School”

prepared by **Garganta, Guienn; Librero, Alyssa Rica; Mesina, Donna May; Villafania, Georgia Victoria**

The data gathered in the study were properly processed and analyzed using appropriate statistical tools and procedures relevant to the objectives of the research.

This certification is issued upon the request of the researcher for whatever legal and academic purposes it may serve.

Issued this 8th day of May, 2026 at Olongapo City, Philippines.


Kristoffer Paul C. Obispo
Data Analyst



APPENDIX F

COMPLIANCE REPORT

Compliance Report

Title : Counting the Uncounted: Exploring the Mathematical Literacy Level Among Kindergarteners at New Cabalan Elementary School.

Panelist : Ms. Anna Theresa Amarille,

Leader : Alyssa Rica Librero

Members : Guienn Garganta, Donna May Mesina, Georgia Victoria Villafania

Comments/ Recommendations	Action Taken	Page Number	Remarks	Signature
- Replace INPUT in Conceptual Framework with what is given in SOP - Edit OUTPUT content	- Replaced INPUT in Conceptual Framework with what is given in SOP - OUTPUT content edited	5	<i>OK</i>	
- Insert the word "intervention" in SOP, - Remove 2.4	- Inserted the word "intervention" in SOP, - Removed 2.4	7	<i>OK</i>	
- Insert "intervention" in SOP, - Remove 2.4	- Question number 2 was advised by the other panelist to be removed. - Table 2 has been renumbered as Table 5.	26 (page 28)	<i>OK</i>	
Change qualitative to quantitative	Changed qualitative to quantitative	27 (page 29)	<i>OK</i>	
Are there studies that can confirm this overall result?	Explanations have been added	28 (page 30)	<i>OK</i>	
Missing table number and interpretation	- Table number and interpretation have been added. - Table 8 has been renumbered as Table 11.	39 (page 41)	<i>OK</i>	



Missing table number and interpretation	- Table number and interpretation have been added. - Table 9 has been renumbered as Table 12.	41 (page 43)	OK	
Missing table number and interpretation	- Table number and interpretation added. - Table 10 has been renumbered as Table 13.	43 (page 45)	OK	
Transfer the content to Statistical treatment	Content transferred to Statistical treatment page.	45 (21)	OK	
Make summary of findings into a paragraph form	Summary of findings was made into a paragraph form.	48	OK	
Conclusions - must be based in SOP	Conclusions were based on the SOP	51 (page 49)	OK	
Insert the word "intervention" to the sentence.	Summary of findings was made into a paragraph form.	50 (page 48 - 49)	OK	



Compliance Report

Title : Counting the Uncounted: Exploring the Mathematical Literacy Level Among Kindergarteners at New Cabalan Elementary School.

Panelist : **Ms. Keanna Dennise David**

Leader : Alyssa Rica Librero

Members : Guienn Gargania, Donna May Mesina, Georgia Victoria Villafania

Comments/ Recommendations	Action Taken	Page Number	Remarks	Signature
Remove question number 1. Move the word table to the left. Unbold and Italicize the description.	Removed question number 1. Adjusted the word table to the left. Unbolded and Italicized the description. Table 1 has been renumbered as Table 4.	24 (page 26)	Complied	
Remove question number 2. Move the word table to the left. Unbold and Italicize the description.	Removed question number 2. Adjusted the word table to the left. Unbolded and Italicized the description. Table 2 has been renumbered as Table 5.	26 (page 28)	Complied	
Move the word table to the left. Unbold and Italicize the description.	Adjusted the word table to the left. Unbolded and Italicized the description. Table 3 has been renumbered as Table 6.	28 (page 30)	Complied	
Move the word table to the left.	Adjusted the word table to the left.	31 (page 33)	Complied	



Unbold and Italicize the description.	Unbolded and Italicized the description. Table 4 has been renumbered as Table 7.		Completed	[REDACTED]
Move the word table to the left. Unbold and Italicize the description.	Adjusted the word table to the left. Unbolded and Italicized the description. Table 5 has been renumbered as Table 8.	33 (page 35)	Completed	[REDACTED]
Move the word table to the left. Unbold and Italicize the description.	Adjusted the word table to the left. Unbolded and Italicized the description. Table 6 has been renumbered as Table 9.	35 (page 37)	Completed	[REDACTED]
Move the word table to the left. Unbold and Italicize the description.	Adjusted the word table to the left. Unbolded and Italicized the description. Table 7 has been renumbered as Table 10.	37 (page 39)	Completed	[REDACTED]
Remove question number 3. Move the word table to the left. Unbold and Italicize the description.	Removed question number 3. Adjusted the word table to the left. Unbolded and Italicized the description. Table 8 has been renumbered as Table 11.	39 (page 41)	Completed	[REDACTED]
Remove question number 4. Move the word table to the left. Unbold and Italicize the description.	Removed question number 4. Adjusted the word table to the left. Unbolded and Italicized the description.	41 (page 43)	Completed	[REDACTED]

Interpretation of
Score pg. 21-23
Table 1, 2, 3 APA format

pg. 21-23



Remove question number 5. Move the word table to the left. Adjust the font size inside the table.	Table 9 has been renumbered as Table 12. Removed question number 5. Adjusted the word table to the left. Adjusted the font size inside the table. Table 10 has been renumbered as Table 13.	43 (page 45)	Complied	
Must be APA format Insert in Appendices Arrange in paragraph form	APA formatted Inserted in Appendices Arranged in paragraph form	45 – 47 (page 79)	Complied	
Unbold "economies"	Unbolded "economies"	48 – 50 (page 48)	Complied	
Not justified Remove Kerdy	Justified Removed Kerdy	56 (page 54)	Complied	
Not justified	Justified	57 (page 55)	Complied	
Move the table to the center Unbold the last phrase	Table moved to the center Unbolded the last phrase	58 (page 56)	Complied	
Remove Thank you very much	Thank you very much removed	79 (page 77)	Complied	
		80 (page 78)	Complied	
		81 (page 81)	Complied	



Compliance Report

Title : Counting the Uncounted: Exploring Mathematical Literacy Level Among Kindergarteners at New Cabalan Elementary School.

Panelist : Ms. Keanna Dennise David

Leader : Alyssa Rica Librero

Members : Guienn Garganta, Donna May Mesina, Georgia Victoria Villafania

Comments/ Recommendations	Action Taken	Page Number	Remarks	Signature
Tables 1, 2, and 3 to be APA formatted.	Tables 1, 2, and 3 was APA formatted.	Pages 21 - 23	complied	



APPENDIX G

RERU PROTOCOL APPROVAL



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
RESEARCH DEVELOPMENT AND COMMUNITY EXTENSION SERVICES
Olongapo City Sports Complex, Donor St., East Tapinac, Olongapo City



Research Ethics Review Unit Protocol Approval

May 5, 2026

Ms. Alyssa Rica Librero
Principal Investigator

Re: Counting the Uncounted: Exploring the Mathematical Literacy Levels
among Kindergarteners in New Cabalan Elementary School
(GC-RDCES-RERU Code: 062/050526/Librero)

Dear Ma'am,

We wish to inform you that the study protocol entitled "**Counting the Uncounted: Exploring the Mathematical Literacy Levels among Kindergarteners in New Cabalan Elementary School**" has been reviewed and is hereby granted **Ethics Clearance** for implementation by the GC-RDCES-RERU. The study has been assigned study protocol code **062/050526/Librero**, which should be used for all communication to the GC-RDCES-RERU related to this study. This ethical clearance is valid for one year until 05/05/2027.

The following technical documents were submitted for review on which this approval was based:

- 1) Study protocol/ Investigator's brochure
- 2) Proposal Manuscript
- 3) A comprehensive review of literature that justified the conduct of the study
- 4) The procedure of research data gathering was found to be scientific
- 5) Adherence and observance of ethical standards set for research undertaking as seen as Chapter 2 – Ethical Consideration.
- 6) Reference List
- 7) Informed Consent Form/ Assent Form
- 8) Appended Research Instrument
- 9) Curriculum Vitae illustrating the Investigator's qualification to conduct of the study
- 10) Budget Proposal using the GC-RDCES-RERU-Form-004 (Detailed Budget Sheet)
- 11) Compliance and justification report on the protocol assessment form

While the study is in progress, the Principal Investigator is hereby requested to submit the following documents:

- 1) Progress report using the GC-RDCES-RERU-Form-006 (Continuing Review Application Form) every six months from the date of approval which includes the following: (Note: *In view of active ethical clearance, this report is mandatory even if the study has not started or is still awaiting release of funds*).

Ethics Clearance – 062/050526/Librero

Page 1 of 2



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



Republic of the Philippines
City of Olongapo

GORDON COLLEGE

RESEARCH DEVELOPMENT AND COMMUNITY EXTENSION SERVICES
Olongapo City Sports Complex, Donor St., East Tapinac, Olongapo City



- a) Date covered by the report (if any)
- b) Protocol summary and status report on the progress of the research
- c) Number of participants accrued
- d) Withdrawal or termination of participants
- e) Complaints on the research since the last GC-RDCES-RERU review
- f) Summary of relevant recent research literature, interim findings, and amendments since the last GC-RDCES-RERU review
- g) Any relevant multi-center research reports
- h) A copy of the accomplished informed consent document
- 2) Any changes in the protocol, especially those that may adversely affect the safety of the participants during the conduct of the study/ trial including changes in personnel, must submitted or reported using the GC-RDCES-RERU-Form-005 (Study Protocol Amendment Submission form).
- 3) Revisions in the informed consent form using the GC-RDCES-RERU-Form-005 (Study Protocol Amendment Submission form).
- 4) Reports of adverse events including from other study sites using the GC-RDCES-RERU-Form-009a (Serious Adverse Event Report).
- 5) Notice of early termination of the study and reasons for such using the GC-RDCES-RERU-Form-007 (Early Termination Report Form).
- 6) Any event which may have ethical significance.
- 7) Any information which is needed by the GC-RDCES-RERU to do an ongoing review.
- 8) Notice of time of completion of the study using GC-RDCES-RERU-Form-008 (Final Report Form).
- 9) Application for renewal of ethical clearance is 90 days before the expiration date of this approval through submission of GC-RDCES-RERU-Form 006 (Continuing Review Application Form).
- 10) Any unreported protocol amendments, deviation, serious adverse events, suspected, unexpected serious adverse events, and other gross violations determined by the RERU, ethics clearance may be revoked after a thorough investigation and determination of RERU.

Thank you.

Respectfully,

JOHN MARK R. ASIO, RN, RM, LPT, DPA
Coordinator
RDCES-RERU



Republic of the Philippines
City of Olongapo
GORDON COLLEGE
College of Education, Arts and Sciences



CURRICULUM VITAE



About Me

Dedicated education student with a unique background in client service and administration, ready to foster safe and engaging learning spaces.

Contact

+63-915-499-5033

alylibrero@gmail.com

2042 Old Cabalan, Olongapo City, Zambales, Philippines

Skills

- Sales Strategies
- Negotiation Skills
- Problem-Solving
- Time Management
- Presentation Skills
- Networking
- Team Collaboration

Language

- English
- Tagalog

ALYSSA RICA LIBRERO

Early Childhood Educator

Education

(2005-2006)

NELLIE E. BROWN ELEMENTARY SCHOOL

(2010-2011)

ST. JOSEPH COLLEGE INC.-OLONGAPO CITY

(2023 -present)

GORDON COLLEGE

Bachelor of Early Childhood Education

Professional Experience

(2016 -2017)

OFFICE SECRETARY

R.A Romero Marketing

- Developed and maintained systematic filing and data management systems (both digital and physical), ensuring accuracy, confidentiality, and easy retrieval of critical documents.
- Assisted in budget tracking and office inventory management, contributing to cost-effective resource allocation and operational planning.

(2017 - 2019)

CALL CENTER REPRESENTATIVE

RAS Contact Solutions

- Managed a high volume of inbound/outbound customer service calls in a dynamic, target-driven environment, consistently meeting and exceeding performance metrics for call quality and resolution rate.
- De-escalated tense situations with dissatisfied clients by employing active listening, empathetic communication, and structured problem-solving, successfully converting conflicts into positive service experiences.



DONNA MAY J. MESINA

Passionate student and dedicated educator committed to lifelong learning and empowering others.

PROFESSIONAL STRENGTHS

- Knowledgeable and experienced from dealing with children in the teaching profession.
- Knowledgeable in Desktop Productivity and Designs.
- Possess strong determination to acquire what is asked, disciplined, truthful, reliable, diligent, and open to new ideas.
- Can work with minimum supervision.

WORK HISTORY

ESL Teacher

2019 July - Present

51talk Philippines

- Teaching non-native English globally.

Spa Therapist

2015 October - 2018 May

Timeless Spa

Dubai International Hotel,

Dubai International Airport Dubai, United Arab Emirates

- Performs facial cleansing
- Performs whole body {Swedish, Lomilomi, Shiatsu, Hot Stone, Thai, Scrub} and seated massage.

OJT Trainee/ Admin Clerk

2011 May - 2011 September

PTT Philippines Trading Corporation

Brand Rex Compound, Argonaut Hi way Boton Area,

SBFZ, Philippines

- Exposed to clerical jobs, basic office work and process.

TRAININGS AND SEMINARS

- Discovering Emotional and Social Learning ECCD Council (Nov. 15, 2024)
- (TESOL) Teaching English to Students of Other Language and (TEYL) Teaching English to Young Learners 140 hours -Teach International English Language School, teachinternational.edu.au (August 14, 2019)

EDUCATION

Bachelor of Early Childhood Education

Gordon College 2023-present

Associate in Computer Programming

Central Luzon 2010-2012

College of Science
and Technology

SKILLS

- Communication
 - English & Filipino
- Time management
- Organization
- Technical skills
- Creativity

PERSONAL PROFILE

Birthdate : 02 March 1981

Birthplace : Bacoar, Cavite

Age : 44

Status : Single

Religion : Christian



CURRICULUM VITAE

I. PERSONAL DATA

Name: Georgia Victoria M. Villafania

E-mail: georgiamontevirgen2@gmail.com

Date of Birth: May 2, 2002

Place of Birth: Olongapo City, Zambales



II. EDUCATIONAL BACKGROUND

Tertiary: Gordon College

Bachelor of Early Childhood Education

Olongapo City Sports Complex, Donor Street, East Tapinac, Olongapo
City, Zambales, 2200

2021 – Present

Senior High School: College of Subic Montessori

Accountancy and Business Management (ABM)

Bldg 737, Labitan St, CBD, SBFZ, 2222 Zambales

2018 – 2020

Junior High School: Olongapo City National High School

Rizal Avenue, East Tapinac, Olongapo City, Zambales, 2200

2014 – 2018

Elementary: Olongapo City National High School

Rizal Avenue, West Bajac Bajac O.C., Olongapo, Philippines

2007 - 2014



GUIENN D. GARGANTA

 #6 Adarna st. Prk. 6, Brgy. New Cabalan, Olongapo, Zambales, 2200
 +63 966 659 7171
 guienndelareagarganta@gmail.com



PERSONAL INFORMATION

- Date of Birth: September 15, 2003
- Gender: Female

EDUCATION

- **Gordon College**
Bachelor of Early Childhood Education
Present - 2027
- **College of Subic Montessori**
ABM Strand
With honors, Best in work Immersion
Graduated year 2021
- **Olongapo City National High School**
Graduated year 2019
- **Tapinac Elementary School**
Graduated year 2015

VOLUNTEER/CERTIFICATES

- **Summer Pre-Kindergarten Program**
Teacher
New Era University Project
From year 2022-2025